

Panel Conditioning in Fixed-Effects Models: Identification and Bias Propagation*

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Abstract

Panel conditioning—the causal effect of prior survey participation on survey responses—is widely documented, and the mean path of conditioning over tenure is the object most of that literature reports. We show what a staggered panel identifies about that path and what the unidentified remainder does to the estimators panel data are used for. First, on the finite support a panel fields, the identified set of the path is the translate of the tenure components of the cell design’s kernel, and a functional of the path is identified exactly when it is orthogonal to that kernel. Because tenure, calendar time, and entry cohort are linearly dependent, the kernel always contains an affine direction and, when the entry cohorts share a common stride d , $d - 1$ periodic directions; under a connectivity condition on observed increments these exhaust it, and the identified set has dimension d . The affine direction is the age–period–cohort dependency and the underidentification of fully dynamic event studies; the periodic directions are the unequal-interval APC ambiguity in the measurement domain. Curvature is then identified only at the design’s stride, so that a panel refreshed every four waves does not identify ordinary second differences, and a lemma gives the follow-up a refreshment cohort needs for the connectivity condition to hold. A lemma bounds the dimension for any number of cohorts and shows that the last refreshment cohort’s follow-up is itself an identification parameter. When participation is interrupted, as in rotation designs, the clock that cannot be seen is calendar time rather than the dose: the path is identified up to an affine function of the elapsed time at which each interview arrives, so that a design whose interviews are unequally spaced—the four-eight-four rotation of the Current Population Survey—distinguishes conditioning that accumulates per interview from conditioning that accumulates per month, which no equally spaced design can. Identifying restrictions—a plateau, entry-wave negative controls, bounded cohort drift—are stated together with what each removes and what remains: negative

*Code reproducing every numerical result in this paper, including the downstream identities of Section 6 and every row of Tables 1 and 2, is in the replication archive at <https://github.com/sokubo/paper-panel-conditioning-replication> (fixed version: tag `paper-v0.7`, commit `a48bd80`). No restricted data are used. This work was supported by JSPS KAKENHI Grant Number 22K13525.

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controls remove the affine direction but not the periodic ones. Second, an absorption theorem establishes that two-way fixed-effects estimators annihilate every kernel direction, so their bias depends on conditioning only through its identified part, is invariant to every normalization, and is itself identified: the two-way regression with tenure indicators added is a corrected estimator, subject to a residual-rank condition. A spurious pre-trends theorem shows that when event time is aligned with tenure, fully dynamic event-study coefficients inherit the non-affine part of the path exactly, producing pre-trends without anticipation that outcome-side diagnostics may detect but cannot attribute, and that cannot be corrected within the sample; a sensitivity bound turns any bound on identified curvature into a bound on every event-study coefficient, without normalizing the level of conditioning. Simulations verify each identity to machine precision and report finite-sample behavior; a 19-wave Japanese panel with two refreshment samples illustrates the design calculus.

Keywords: panel conditioning; identification; age-period-cohort; two-way fixed effects; event study; refreshment samples

1 Introduction

Panel surveys ask the same individuals the same questions repeatedly, and a long literature documents that the act of being interviewed changes later answers—*panel conditioning* (Warren and Halpern-Manners 2012; Halpern-Manners and Warren 2012; Halpern-Manners, Warren, and Torche 2017). Formal identification analysis of the phenomenon is rare: Das, Toepoel, and van Soest (2011) give one for a binary item in a two-wave design with a refreshment sample, and Feng, Hu, and Sun (2022) give one for a latent-state model of labor-force misclassification in the Current Population Survey. For the object that most of the applied literature reports—a mean path of conditioning over tenure, estimated from the cell means of a staggered panel—practitioners have long behaved as if a non-identification result existed: the U.S. Bureau of Labor Statistics normalizes month-in-sample effects to average to one; Krueger, Mas, and Niu (2017) subtitle their study of rotation group bias “Will the Real Unemployment Rate Please Stand Up?” precisely because, as they put it, “ex ante it is unclear which of the CPS rotation groups has less bias”; van den Brakel and Krieg (2015) estimate rotation-group bias only relative to a reference group. Each of these choices is a normalization. This paper states the result that forces it, proves it on the finite support a panel actually fields, characterizes what it leaves identified—which depends on the refreshment schedule and on how long each cohort is followed—and derives what the unidentified

component does to the fixed-effects and event-study estimators that panel data are used for. The first half concerns identification. Theorem 1 states what a panel identifies about the conditioning path $\tau(\cdot)$ on the finite support it actually fields: the identified set is the translate of the tenure components of the cell design’s kernel, and a functional of τ is identified exactly when it is orthogonal to that kernel. Two kinds of direction are in the kernel of every staggered design. The affine direction is there because tenure s , calendar time t , and entry cohort e satisfy $s = t - e + 1$; it is the exact translation into the measurement domain of the age–period–cohort linear dependency (Mason et al. 1973; Fosse and Winship 2019a) and of the underidentification of fully dynamic event-study designs (Borusyak, Jaravel, and Spiess 2024). When the entry cohorts share a common stride d —the greatest common divisor of their spacings— $d - 1$ periodic directions are there as well; these are the measurement-domain form of the additional non-identifiability of age–period–cohort models with unequal intervals (Gascoigne and Smith 2023), and they bite: a panel refreshed at a four-wave stride has an identified set of dimension at least four, and the Japanese panel we use for illustration is one. Under a connectivity condition on the panel’s observed increments the affine and periodic directions are all there is, and Lemma 1 says when a staggered panel satisfies it—the last refreshment cohort must be followed for enough periods—and gives the exact dimension for three cohorts. Corollary 1 characterizes the identified functionals—curvature at the design’s stride, ordinary curvature only when entry cohorts are adjacent. Section 3.1 then removes the assumption that participation is uninterrupted, which rotation designs violate by construction, and finds that the relaxation carries identifying power: the unidentified direction is an affine function of the *elapsed calendar time* at which each interview arrives, not of the interview count, so that a panel whose interviews are unequally spaced can separate conditioning that accumulates per interview from conditioning that accumulates per month—the Current Population Survey’s four-eight-four rotation is such a design, and Corollary 2 exhibits the contrast, estimable from published month-in-sample means, that answers the question. Section 5 gives identifying restrictions (a plateau, entry-wave negative controls, bounded cohort drift), each stated together with what it removes and what is and is not testable about it: negative controls remove the affine direction but not the periodic ones, and zero identified curvature is consistent with a linear path, so saturation cannot be established

from cell means alone.

The second half turns from what cannot be known to where it goes. Applied panel analysis treats measurement error as stationary; the conditioning literature says it is not. Section 6 derives the consequences for the estimators panel users actually run and finds them exactly complementary to Theorem 1. The *absorption theorem* (Theorem 3) shows that two-way fixed-effects estimators annihilate precisely the unidentified component of the conditioning path—every kernel direction’s tenure component is a sum of a time function and a unit function, on any support—so that the bias of a fixed-effects coefficient depends on conditioning only through its identified part, is invariant to the normalization every conditioning study must choose, and is itself an identified functional (Theorem 4). A feasible corrected estimator is the two-way regression with tenure indicators added, subject to a residual-rank condition that fails when exposure is scheduled by tenure. The *spurious pre-trends theorem* (Theorem 5) locates the boundary of that protection: when event time is aligned with tenure, fully dynamic event-study coefficients absorb the non-affine part of the path exactly, so that pre-period coefficients are nonzero without any anticipation or violation of parallel trends in the outcome, an artifact that outcome-side diagnostics (Roth 2022; Rambachan and Roth 2023) may detect but cannot attribute, that cannot be corrected within the aligned sample, and that an external estimate of the conditioning path can remove. Because that external estimate is rarely available as a point, Proposition 5 supplies the sensitivity version: a bound on the identified curvature of the path implies an explicit bound, growing quadratically in the distance from the reference categories, on the measurement-induced movement of every event-study coefficient—and it needs no normalization of the level of conditioning, because the movement it bounds is invariant to the unidentified affine direction. An orthogonality characterization (Proposition 6) states in one line which linear panel functionals survive which classes of drift.

Relation to Feng, Hu, and Sun (2022). Their result achieves *point* identification of CPS misclassification probabilities by restricting response dynamics to first-order dependence on the previous report, an exclusion restriction linking a later interview to the latent status of an earlier one, rank and eigenvalue conditions, and a cross-rotation-group comparability

condition under which the estimated error probabilities are transported to the remaining groups. The model class is different from ours—latent states and misclassification matrices rather than an additive mean path—and we do not claim that their restrictions select a point of our identified set. What the two settings share is the dependency between tenure, period, and cohort; our entry-wave negative controls check one component of their cross-rotation-group comparability condition—that entering rotation groups are comparable on time-invariant facts—and not the full latent-state and response-history transport that their result requires.

2 Setup and Notation

Individuals i belong to entry cohort $e_i \in \mathcal{E} \subset \mathbb{Z}$ and are eligible for interviews at calendar times $t \in \mathcal{T} \subset \mathbb{Z}$. Tenure at time t is

$$s_{it} = t - e_i + 1 \geq 1.$$

Let $Y_{ijt}(s)$ denote the potential response of individual i to item j at time t when the measurement history is coarsened to tenure s (Assumption M1 below). Observed responses are $Y_{ijt} = Y_{ijt}(s_{it})$ for respondents. We suppress j where possible. Define cell means over the target population

$$\mu(e, t) = \mathbb{E}[Y_{it}(t - e + 1) | e_i = e], \quad (e, t) \in \mathcal{S},$$

where $\mathcal{S} \subset \mathcal{E} \times \mathcal{T}$ is the *observed support*: the finite set of (cohort, period) cells the panel fields. The leading case is the staggered trapezoid $\{(e, t) : e \leq t \leq T\}$, every cohort followed from entry to a common final period T , but nothing below requires it. Write $\mathcal{S}_{\text{obs}} = \{t - e + 1 : (e, t) \in \mathcal{S}\}$ for the set of observed tenures and $\mathcal{T}_{\text{obs}}$ for the set of observed periods.

We maintain throughout this section (and relax later):

- **M1 (coarsening / consistency).** Potential responses depend on the measurement

history only through tenure: $Y_{ijt}(h) = Y_{ijt}(|h| + 1)$, and all cohorts are measured with the same instrument and mode at a given t . Participation is uninterrupted, so that calendar tenure $t - e + 1$ equals the number of interviews received. In interrupted rotation designs—the CPS 4–8–4 pattern, in which a household is interviewed for four consecutive months, rested for eight, and returned for four more—the two differ, and the count of interviews, not the calendar spacing, is the argument of τ ; Section 3.1 treats that case, and the difference turns out to carry identifying power rather than to be a technicality.

- **M2 (sampling equivalence).** Cohorts are probability samples from the same target population, so that cohort differences in μ reflect entry-cohort effects and conditioning only.
- **M3 (no attrition).** All entrants respond at all $t \geq e$. (Relaxed in Section 4.)
- **M4 (additive cell means).** On the observed support,

$$\mu(e, t) = \alpha(t) + g(e) + \tau(s), \quad s = t - e + 1, \quad (1)$$

with the normalizations $\tau(1) = 0$ (no conditioning at first interview, by definition) and $g(e_0) = 0$ for a reference cohort e_0 .

M4 is an assumption about the response process, not a consequence of M1–M3: it says that period, cohort, and conditioning enter cell means additively, with no cohort-by-tenure or period-by-tenure interaction. It is the standard decomposition of the rotation-group-bias and panel-conditioning literatures, and everything below is a statement about the additive model; with interactions, the indeterminacy is item-wise or cohort-wise and larger. The object of interest is the conditioning path τ restricted to the observed tenures, $\tau : \mathcal{S}_{\text{obs}} \rightarrow \mathbb{R}$; values of τ at tenures the panel never observes are outside the scope of any identification statement.

3 The Impossibility Theorem

Let the entry cohorts be $\mathcal{E} = \{e_1 < \dots < e_K\}$, $K \geq 2$, with reference cohort $e_0 = e_1$, and define the design's *stride*

$$d := \gcd\{e_k - e_j : e_k, e_j \in \mathcal{E}\}.$$

Consecutive entry cohorts give $d = 1$; a panel refreshed every four waves, or with refreshments four and twelve waves after the first entry, gives $d = 4$. A function $\rho : \mathbb{Z} \rightarrow \mathbb{R}$ is *d-periodic* if $\rho(s + d) = \rho(s)$ for all s ; when $d = 1$ the only d -periodic function with $\rho(1) = 0$ is $\rho \equiv 0$.

The design matrix and its kernel. Stack the cell equations Equation 1 over $(e, t) \in \mathcal{S}$ as $\mu = X_{\mathcal{S}} \theta$, where $\theta = (\alpha, g, \tau)$ collects the free parameters— $\alpha(t)$ for $t \in \mathcal{T}_{\text{obs}}$, $g(e)$ for $e \neq e_0$, $\tau(s)$ for $s \in \mathcal{S}_{\text{obs}} \setminus \{1\}$ —and $X_{\mathcal{S}}$ is the $|\mathcal{S}| \times (\text{number of parameters})$ incidence matrix with a one in the columns of $\alpha(t)$, $g(e)$, and $\tau(t - e + 1)$ of each fielded cell. Two parameter vectors generate the same cell means if and only if their difference lies in the kernel $\mathcal{K} := \ker X_{\mathcal{S}}$. Write $h = (h_{\alpha}, h_g, h_{\tau})$ for an element of $\mathcal{K}$ and $\mathcal{K}_{\tau} := \{h_{\tau} : h \in \mathcal{K}\}$ for the set of its tenure components. Everything the panel can and cannot reveal about the conditioning path is a statement about $\mathcal{K}_{\tau}$.

Increments and connectivity. An *observed increment* is a tenure u such that some cohort is observed at tenures u and $u + 1$ in consecutive periods: $(e, t), (e, t + 1) \in \mathcal{S}$ with $u = t - e + 1$. Let $\mathcal{U}$ be the set of observed increments. The *increment graph* has vertex set $\mathcal{U}$ and an edge between u and u' whenever two cohorts $e \neq e'$ are both observed at some t and $t + 1$ with tenures $u = t - e + 1$ and $u' = t - e' + 1$. Every edge joins increments whose difference $e' - e$ is a multiple of d , so the connected components of the increment graph are contained in residue classes modulo d . We say that $\mathcal{S}$ satisfies condition $\mathbf{C}_d$ (*tenure-connectivity at stride d*) if (i) it is *increment-complete*: for every observed tenure s , every $u \in \{1, \dots, s - 1\}$ is an observed increment, so that each observed tenure is reached from tenure one through observed increments; and (ii) the connected components of the increment graph are exactly the residue classes of $\mathcal{U}$ modulo d . The staggered trapezoid is increment-complete because its first cohort is observed in every period; (ii) is the substantive condition, and Lemma 1

below says when a trapezoid satisfies it. The vertices of the graph are increments, not tenure levels: the largest observed tenure is never an increment, and a graph drawn on tenure levels would leave it isolated.

Theorem 1 (Non-identification of the conditioning path). *Under M1–M4 and the normalizations $\tau(1) = 0$, $g(e_0) = 0$, on any finite observed support $\mathcal{S}$:*

(a) (Identified set.) *The identified set for the conditioning path is the affine family*

$$\Theta_\tau = \tau + \mathcal{K}_\tau = \{ \tau + h_\tau : h \in \ker X_{\mathcal{S}} \},$$

and a linear functional $\lambda'\tau = \sum_s \lambda_s \tau(s)$ is point identified if and only if $\lambda' h_\tau = 0$ for every $h_\tau \in \mathcal{K}_\tau$.

(b) (Directions that are never identified.) *For every $m \in \mathbb{R}$ and every d -periodic ρ with $\rho(1) = 0$, the vector*

$$h_\tau(s) = m(s-1) + \rho(s), \quad h_g(e) = m(e-e_0), \quad h_\alpha(t) = -m(t-e_0) - \rho(t-e_0+1)$$

lies in $\ker X_{\mathcal{S}}$. These vectors form a linear space $\mathcal{K}_0 \subseteq \mathcal{K}$ —one affine direction and $d-1$ periodic directions—whose dimension is d whenever $\mathcal{S}$ contains d consecutive observed periods, as every staggered trapezoid with $T \geq e_1 + d-1$ does. Consequently $\dim \Theta_\tau \geq d$ on such supports; no functional with $\sum_s \lambda_s(s-1) \neq 0$ is identified on any support; and when $d > 1$ neither is any functional whose weights fail to sum to zero within some residue class modulo d other than that of $s = 1$.

(c) (Sharpness under C_d .) *If $\mathcal{S}$ satisfies C_d , then $\mathcal{K}_\tau$ is exactly the set of tenure components of $\mathcal{K}_0$:*

$$\Theta_\tau = \{ s \mapsto \tau(s) + m(s-1) + \rho(s) : m \in \mathbb{R}, \rho \text{ } d\text{-periodic}, \rho(1) = 0 \},$$

a d -dimensional affine family; and $\lambda'\tau$ is identified if and only if $\sum_s \lambda_s(s-1) = 0$ and $\sum_{s \equiv r \pmod{d}} \lambda_s = 0$ for every residue $r \not\equiv 1$. In particular, if $d = 1$ and C_1 holds, $\Theta_\tau = \{ \tau + m(s-1) \}$: the level and linear trend of conditioning are not identified and

nothing else is lost.

(d) (Design matrix.) $\dim \Theta_\tau = \dim \mathcal{K}_\tau$, and if the cohort-period incidence graph of $\mathcal{S}$ is connected (as it is for every staggered panel), $\dim \mathcal{K} = \dim \mathcal{K}_\tau$ and $\text{rank } X_\mathcal{S} = (\text{number of parameters}) - \dim \Theta_\tau$. The rank is at most the number of parameters minus d , with equality under C_d . A design's identified space—its dimension and a basis of $\mathcal{K}_\tau$ —can therefore be computed, before any data are collected, from $X_\mathcal{S}$ alone; C_d is a sufficient condition for it to have the closed form in (c), not a prerequisite for computing it.

Proof. (a) Two parameter vectors θ, θ' generate the same cell means iff $X_\mathcal{S}(\theta' - \theta) = 0$, i.e., $\theta' - \theta \in \mathcal{K}$; restricting to the τ -coordinates gives $\Theta_\tau = \tau + \mathcal{K}_\tau$. A linear functional is constant on an affine family iff it annihilates the family's direction space.

(b) At any $(e, t) \in \mathcal{S}$ with $s = t - e + 1$,

$$h_\alpha(t) + h_g(e) + h_\tau(s) = m[-(t - e_0) + (e - e_0) + (s - 1)] + [\rho(t - e + 1) - \rho(t - e_0 + 1)] = 0,$$

because $-(t - e_0) + (e - e_0) + (t - e + 1 - 1) = 0$ and because $e \equiv e_0 \pmod{d}$ for every $e \in \mathcal{E}$ (d divides $e - e_0$), so that $\rho(t - e + 1) = \rho(t - e_0 + 1)$. The normalizations are respected: $h_\tau(1) = \rho(1) = 0$ and $h_g(e_0) = 0$. The map $(m, \rho) \mapsto h$ is linear; it is injective when d consecutive periods are observed, because h_g at any $e \neq e_0$ determines m and then $h_\alpha(t) = -m(t - e_0) - \rho(t - e_0 + 1)$ over d consecutive periods determines the d -periodic ρ ; and the space of d -periodic ρ with $\rho(1) = 0$ has dimension $d - 1$. The two functional statements follow from (a) by taking $\rho = 0$, and $m = 0$ with ρ the indicator of one residue class $r \neq 1$.

(c) Let $h = (h_\alpha, h_g, h_\tau) \in \mathcal{K}$, so $h_\alpha(t) + h_g(e) + h_\tau(t - e + 1) = 0$ on $\mathcal{S}$, $h_\tau(1) = 0$, $h_g(e_0) = 0$. Whenever (e, t) and $(e, t + 1)$ are observed, differencing gives $\Delta h_\alpha(t) = -\Delta h_\tau(u)$ with $u = t - e + 1$ and $\Delta f(x) := f(x + 1) - f(x)$. The left side does not depend on e , so whenever two cohorts $e \neq e'$ are observed at t and $t + 1$ with increments u, u' , $\Delta h_\tau(u) = \Delta h_\tau(u')$: the first difference of h_τ is constant along every edge of the increment graph, hence constant on each of its components, hence, by $C_d(\text{ii})$, constant on each residue class of $\mathcal{U}$ modulo

d. Write $\Delta h_\tau(u) = c_{[u]}$ with $[u]$ the residue of u , let $m := d^{-1} \sum_r c_r$ be the mean over one period (with $c_r := 0$ for any residue class not represented in $\mathcal{U}$), and set $\rho(s) := \sum_{u=1}^{s-1} c_{[u]} - m(s-1)$ for every integer $s \geq 1$, $\rho(1) = 0$. Then $\Delta\rho(s) = c_{[s]} - m$ is d -periodic with zero mean over a period, so $\rho(s+d) - \rho(s) = 0$: ρ is d -periodic. By increment-completeness, for every observed tenure s the increments $1, \dots, s-1$ are all observed, so $h_\tau(s) = h_\tau(1) + \sum_{u=1}^{s-1} \Delta h_\tau(u) = m(s-1) + \rho(s)$. Back-substituting, $h_\alpha(t) + h_g(e) = -m(t-e) - \rho(t-e+1) = -m(t-e_0) + m(e-e_0) - \rho(t-e_0+1)$, whose right side is a function of t plus a function of e ; hence $h_\alpha(t) = -m(t-e_0) - \rho(t-e_0+1) + \kappa$ and $h_g(e) = m(e-e_0) - \kappa$ on each connected component of the cohort-period incidence graph, and $h_g(e_0) = 0$ forces $\kappa = 0$ on the component containing e_0 . Thus h_τ is the tenure component of an element of $\mathcal{K}_0$, which gives $\mathcal{K}_\tau = \{m(s-1) + \rho(s)\}$ and, with (a), the two displayed statements; the functional characterization is the case of (a) in which $\mathcal{K}_\tau$ is spanned by $s-1$ and the residue-class indicators $r \neq 1$. The $d=1$ clause is the special case $\rho \equiv 0$.

(*d*) $\dim \Theta_\tau = \dim \mathcal{K}_\tau$ by (a). If the incidence graph is connected, the argument at the end of (c) shows that h_τ determines (h_α, h_g) up to a constant κ that the normalization $h_g(e_0) = 0$ fixes, so the projection $h \mapsto h_\tau$ is injective on $\mathcal{K}$ and $\dim \mathcal{K} = \dim \mathcal{K}_\tau$; the rank statement is the rank-nullity theorem. The inequality is (b) and the equality under C_d is (c). $\square \quad \square$

The theorem separates three things that an earlier formulation of ours ran together. Part (a) is the general statement: on a finite support, the identified set is the translate of the kernel's tenure components, and identification of a functional is orthogonality to that kernel—a computation, valid for any fielded support including ones with gaps or unequal follow-up. Part (b) says which directions are in the kernel of *every* design with the given entry cohorts: the affine direction, and—when the cohorts share a stride—the periodic ones. Part (c) says when those are *all* there is. The distinction matters because C_d can fail on ordinary designs: with cohorts entering at 1, 4, and 8 and observation through period 8, the stride is $d=1$, yet the 14-cell design has 17 parameters and rank 14, so the identified set is three-dimensional, not one-dimensional—the last cohort is observed only at entry, its increments merge nothing, and the increment graph has the three components $\{1, 4, 7\}$, $\{2, 5\}$, $\{3, 6\}$.

Part (a) covers such a design exactly; part (c) does not apply to it. The Japanese panel of Section 8, with cohorts entering at waves 1, 5, and 13 and observed through wave 19, has $d = 4$, satisfies C_4 , and has a design matrix with 41 cells, 39 free parameters, and rank 35: its identified set is four-dimensional. The support $e \in \{1, 3\}$, $t \in \{3, 4, 5\}$ with the perturbation $h_\tau(s) = \mathbf{1}\{s \text{ even}\}$, $h_\alpha(t) = -\mathbf{1}\{t \text{ even}\}$, $h_g \equiv 0$ is the smallest example of a periodic direction: it preserves every cell mean and both normalizations, is not affine, and has $\Delta^2 h_\tau(2) = -2$.

Antecedents. Neither kind of direction is new to the identification literature, and we position the theorem accordingly. The affine direction is the exact translation into the measurement domain of the age–period–cohort linear dependency (Mason et al. 1973; Fosse and Winship 2019a) and of the underidentification of fully dynamic event-study specifications without never-treated units (Borusyak, Jaravel, and Spiess 2024; Sun and Abraham 2021), with tenure in the role of age or event time. The periodic directions are the measurement-domain form of the additional non-identifiability of APC models with *unequal intervals*. When age is recorded in intervals M times as wide as periods, so that cohort is $c = p - Ma$, Gascoigne and Smith (2023, sec. 4.1, eqs. (4)–(5)) show that an M -periodic function v_M can be added to the period effect and subtracted from the cohort effect without changing the linear predictor, and that the period and cohort *curvatures*—identified in the equal-interval case—are then unidentified, a phenomenon they call the curvature identifiability problem. Under the correspondence $p = t$, $Ma = e - e_0$, $c = s - 1$, and $M = d$, their v_M is our ρ up to sign: entry cohorts that share a stride are age groups recorded at a coarser interval than periods, and our Theorem 1(b) is their equation (4) with tenure in the role of cohort. The problem is older than that paper, which credits Holford (2006) with first identifying it—as the “micro-trend identifiability problem,” in factor models fitted to unequally aggregated APC data—and reports earlier representations of the same periodicity by an indicator function (Smith and Wakefield 2016) and by a modulo function (Riebler and Held 2010); we have read Gascoigne and Smith (2023) in the published version and take the attribution of the three earlier works from it. Borusyak, Jaravel, and Spiess (2024, sec. 3.2, fn. 15) note in passing that additional collinearity arises in event studies when treatment is staggered at periodic intervals. What the present theorem adds to these is the finite-support formulation (a)–(d)—

the exact identified set on the support a panel actually fields, the increment-graph condition under which the affine-plus-periodic family is exhaustive, the functional characterization at the design's stride, and the rank computation—together with Lemma 1, which gives the follow-up condition and the exact dimension for the staggered trapezoid, and the downstream results of Section 6, which say what the unidentified component does to fixed-effects and event-study estimators. We make no claim to have discovered periodic non-identification.

Lemma 1 (Staggered trapezoids). *Let $\mathcal{S} = \{(e, t) : e \in \mathcal{E}, e \leq t \leq T\}$ with $e_1 < \dots < e_K$, $K \geq 2$, stride d , and write $\Delta_2 := e_2 - e_1$ for the spacing between the first two cohorts, $w := T - e_K$ for the number of periods, beyond its entry period, during which the last cohort is observed, and $\nu(\mathcal{S}) := \dim \Theta_\tau$.*

- (o) (Component count.) *$\nu(\mathcal{S})$ equals the number of connected components of the increment graph on $\mathcal{U} = \{1, \dots, T - e_1\}$. Hence $\nu(\mathcal{S}) \geq d$, with equality if and only if $\mathcal{S}$ satisfies C_d .*
- (p) (Sufficient condition.) *If $w \geq \Delta_2 - d$, then $\mathcal{S}$ satisfies C_d and $\nu(\mathcal{S}) = d$. In particular, when the first two cohorts are d apart, every $T \geq e_K$ suffices.*
- (ii) (Three cohorts.) *If $K = 3$, then for every $T \geq e_3$,*

$$\nu(\mathcal{S}) = \max\{d, \Delta_2 - w\},$$

so the condition in (i) is necessary as well as sufficient: a third cohort observed for w periods removes exactly w dimensions from the two-cohort indeterminacy Δ_2 , down to the floor d .

- (iii) (Two cohorts.) *If $K = 2$, $\nu(\mathcal{S}) = \Delta_2 = d$ for every $T \geq e_2$.*
- (iv) (Upper bound for any number of cohorts.) *For every $K \geq 3$ and every $T \geq e_K$,*

$$\nu(\mathcal{S}) \leq \max\{d, \Delta_2 - w\},$$

so the three-cohort formula is an upper bound in general: extra cohorts can only shrink the identified set, and the follow-up of the last cohort is what the bound is stated in.

Proof. (o) Let $h \in \mathcal{K}$. The trapezoid is increment-complete, and the first cohort is observed at every $t \in [e_1, T]$, so h_τ is determined on the whole observed tenure range by $h_\tau(1) = 0$ and the first differences $\Delta h_\tau(u)$, $u \in \mathcal{U}$, and by the proof of Theorem 1(c) these differences are constant on the components of the increment graph. Conversely, let Δh_τ be any function on $\mathcal{U}$ that is constant on components, define h_τ by summation, and set $f(t, e) := -h_\tau(t - e + 1)$. The system $h_\alpha(t) + h_g(e) = f(t, e)$ on $\mathcal{S}$ is solvable: put $h_g(e) := f(T, e) - f(T, e_1)$ and $h_\alpha(t) := f(t, e_1)$; the residual $f(t, e) - h_\alpha(t) - h_g(e) = f(t, e) - f(t, e_1) - f(T, e) + f(T, e_1)$ is a rectangle contrast between cohorts e and e_1 over periods t and T , both cohorts being observed throughout $[t, T]$, so it is the telescoping sum over $t' \in [t, T - 1]$ of the consecutive-period contrasts $\Delta f(t', e) - \Delta f(t', e_1) = \Delta h_\tau(t' - e_1 + 1) - \Delta h_\tau(t' - e + 1)$, each of which vanishes because $t' - e + 1$ and $t' - e_1 + 1$ are joined by an edge. Hence every assignment of one free value per component extends to an element of $\mathcal{K}$, uniquely once $h_g(e_0) = 0$ is imposed, and $\nu(\mathcal{S})$ is the number of components. Components lie inside residue classes modulo d , so there are at least d of them when every class is represented in $\mathcal{U}$ (which holds since $\mathcal{U} \supseteq \{1, \dots, \Delta_2\}$ and $d \leq \Delta_2$), with equality iff the components are the residue classes, which is $C_d(\text{ii})$.

Seeds. The pair (e_1, e_k) contributes the edges $u \sim u + \Delta_k$, $\Delta_k := e_k - e_1$, for every $u \in [1, T - e_k]$; edges between two later cohorts $e_j < e_k$ at the same t join $t - e_k + 1$ and $t - e_j + 1$, both of which are joined to $t - e_1 + 1$, so they add no components. The pair (e_1, e_2) joins u to $u + \Delta_2$ for all $u \leq T - e_2$, i.e., whenever both lie in $\mathcal{U}$; hence every $u \in \mathcal{U}$ is in the component of its seed $x \in [1, \Delta_2]$, $x \equiv u \pmod{\Delta_2}$, and $\nu(\mathcal{S})$ is the number of components of the graph induced on the seeds. With $K = 2$ the seeds are pairwise disconnected and $\nu = \Delta_2 = d$: (iii). For $K \geq 3$, the pair (e_1, e_k) joins the seed of u to the seed of $u + \Delta_k$ for each $u \leq T - e_k$; on the seed set $\mathbb{Z}_{\Delta_2} = \{1, \dots, \Delta_2\}$ this is the merge $x \sim x + \Delta_k \pmod{\Delta_2}$, available for every seed $x \leq T - e_k$, hence for every $x \leq w$ (since $T - e_k \geq T - e_K = w$).

(i). The subgroup of $\mathbb{Z}_{\Delta_2}$ generated by $\{\Delta_k \bmod \Delta_2 : k \geq 2\}$ is $d\mathbb{Z}_{\Delta_2}$, because $\gcd(\Delta_2, \dots, \Delta_K) = d$; its cosets are the residue classes modulo d , each of size Δ_2/d . Consider one class C and the directed graph on C with edges $x \rightarrow x + \Delta_k$ for every k and every $x \in C$ with $x \leq w$; vertices $x > w$ have no out-edges. If $w \geq \Delta_2 - d$, the interval

$(w, \Delta_2]$ has length at most d and therefore contains at most one element of C : at most one vertex of C lacks out-edges. Let A be a weakly connected component of this graph. If A contains no vertex without out-edges, then A is closed under $x \mapsto x + \Delta_k$ for every k , hence under addition by the subgroup they generate, hence $A = C$. So every component contains the (at most one) vertex without out-edges, and there is exactly one component: all seeds in C are merged. This holds for every class, so the seeds collapse to d classes, the components of the increment graph are the residue classes, C_d holds, and $\nu = d$ by (o). If $\Delta_2 = d$ the condition $w \geq 0$ is vacuous.

(ii). With $K = 3$ the only merges are $x \sim x + \Delta_3 \pmod{\Delta_2}$ for $x \in [1, w]$: at most w of them, so $\nu \geq \Delta_2 - w$, and $\nu \geq d$ always by (o). If $w \geq \Delta_2 - d$, (i) gives $\nu = d = \max\{d, \Delta_2 - w\}$. If $w < \Delta_2 - d$, the w merges are independent: a cycle among them would require every source in some orbit of $x \mapsto x + \Delta_3$ on $\mathbb{Z}_{\Delta_2}$ —an orbit is a full residue class modulo d within $[1, \Delta_2]$, whose largest element is at least $\Delta_2 - d + 1 > w$ —to lie in $[1, w]$, which is impossible. Hence $\nu = \Delta_2 - w = \max\{d, \Delta_2 - w\}$.

(iv). Every generator Δ_k , $k \geq 3$, is available at every seed $x \leq T - e_k$, hence at every seed $x \leq w$, since $T - e_k \geq T - e_K = w$. Fix a residue class C modulo d within $\{1, \dots, \Delta_2\}$, let $A := C \cap [1, w]$ be its seeds with out-edges and $B := C \setminus A$ those without. If a connected component P of the seed graph restricted to C contains no vertex of B , then every $x \in P$ has all its out-edges, so P is closed under $x \mapsto x + \Delta_k$ for every k , hence under the subgroup these generate—which is $d\mathbb{Z}_{\Delta_2}$, acting transitively on C —so $P = C$ and $B = \emptyset$. Therefore, when $B \neq \emptyset$, every component meets B and distinct components meet disjoint parts of it: the number of components of C is at most $|B| = |C| - |A|$, and it is 1 when $B = \emptyset$. Summing over the d classes, $\nu \leq \sum_C \max\{1, |C| - |A_C|\}$. The classes interleave, so the $|A_C|$ differ by at most one. If some class has $B = \emptyset$ then every class has $|C| - |A_C| \leq 1$, so $\Delta_2 - w \leq d$ and the sum is at most d ; otherwise every term is $|C| - |A_C|$ and the sum is exactly $\Delta_2 - w$. Either way $\nu \leq \max\{d, \Delta_2 - w\}$. $\square$

Three remarks. First, (o) turns C_d into a computation for any trapezoid—count the components of the increment graph—and (i) into a design rule: the last refreshment cohort must be followed for at least $\Delta_2 - d$ periods beyond its entry. In the Japanese panel of Section 8,

$\Delta_2 = 4 = d$ and (i) holds for every T ; had the second cohort entered at wave 6 rather than 5 ($\Delta_2 = 5$, $d = 1$ with the third at 13), the last cohort would have needed $w \geq 4$ periods of follow-up before the identified set shrank to the affine line, and with $T = 16$ ($w = 3$) the identified set is two-dimensional (31 cells, 33 parameters, rank 31). The design with entries at 1, 4, 8 and $T = 8$ discussed after Theorem 1 has $w = 0 < \Delta_2 - d = 2$ and $\nu = 3$, as (ii) gives. Second, (iv) says that the three-cohort formula bounds every design: a fourth or fifth cohort can only merge more seeds, never fewer. The bound is not tight—in the sweep reported in the replication package (all entry sets $\mathcal{E} \subset \{1, \dots, 13\}$ with $3 \leq K \leq 5$ and $T \leq e_K + 13$; 10,934 designs) it is attained in 9,746 of them and strict in 1,188, while (o) holds in every case—so the component count, or the rank of $X_{\mathcal{S}}$, remains the operational check. Stating the bound in terms of the *last* cohort’s follow-up is not an artifact: with entries at 1, 5, 9, 11, 12 and $T = 12$ the third cohort is followed for three periods and the bound computed from it would be 1, while $\nu = 3$, because the spacing $e_3 - e_1 = 8$ is a multiple of $\Delta_2 = 4$ and merges nothing. Third, the replication package provides an R function, `design_rank()`, that builds $X_{\mathcal{S}}$ for any support (trapezoid or not), reports its rank, nullity, stride, and increment-graph components, checks the condition of (i), returns a basis of $\mathcal{K}_{\tau}$, and tests whether a given linear functional of τ is identified.

Remark. Normalizations in practice. Because levels of τ are never identified, every published level of panel conditioning or rotation-group bias rests on a normalization, and the practices of the literature can be read as such: the rotation-group index of the Current Population Survey, defined in the survey’s own documentation as “the ratio of the [estimate] based on a particular month-in-sample group to the average estimate from all eight month-in-sample groups combined, multiplied by 100” (quoted in Halpern-Manners and Warren 2012, fig. 1; the same index is defined and tabulated by Bailer 1975), which sets the average of the eight month-in-sample effects to a constant; Krueger, Mas, and Niu (2017)’s indexation of rotation-group bias to the full-sample mean; the reference-group restriction in the structural time-series model of van den Brakel and Krieg (2015), whose rotation-group bias varies over time and is measured relative to one group; the “average multiplicative bias by month in sample ... relative to average second-stage estimates” tabulated by McIllece (2022, Table 2), which normalizes instead by an external benchmark taken to be unbiased; and the omission

of two reference categories in fully dynamic event studies (Borusyak, Jaravel, and Spiess 2024). We stress that these are analogies between *normalizations*, not equivalences between *models*: the models of van den Brakel and Krieg (2015) and McIllece (2022) are not the time-invariant additive model M4, their estimands differ from τ , and a single restriction of the kind each imposes removes one dimension of Θ_τ , which is all of it only when $d = 1$. The observation that disagreements among published level estimates partly reflect differing conventions is, in our model, a corollary of Theorem 1; it is not a demonstration that those disagreements are exactly movements along Θ_τ , which would require the sources to share M4. A different route is to change the model class rather than to normalize: Feng, Hu, and Sun (2022) point-identify misclassification probabilities in a latent-state model of labor-force status with history-dependent errors by restricting response dynamics to first-order dependence, a model in which the object identified is not a mean path on tenure; we do not claim that their restrictions select a point of Θ_τ , only that the two approaches respond to the same dependency between tenure, period, and cohort.

Corollary 1 (What is identified: curvature at the design’s stride). *On any support, a linear functional $\lambda'\tau$ is point identified if and only if $\lambda \perp \mathcal{K}_\tau$ (Theorem 1(a)). Under C_d this is the pair of conditions*

$$\sum_s \lambda_s (s - 1) = 0 \quad \text{and} \quad \sum_{s \equiv r \pmod{d}} \lambda_s = 0 \quad \text{for every residue } r \not\equiv 1,$$

and in particular the lag- d second differences

$$\Delta_d^2 \tau(s) := \tau(s + 2d) - 2\tau(s + d) + \tau(s)$$

are identified for every s with $s + 2d$ in the support; when $d = 1$ these are the ordinary second differences, and all curvature of τ is identified. When $d > 1$, ordinary second differences $\Delta^2 \tau(s)$ are not identified, and neither are the increments $\tau(s + d) - \tau(s)$ between refreshment tenures: curvature is identified only at the design’s stride. On a support that violates C_d the identified functionals are fewer still, and must be read off $\mathcal{K}_\tau$. Zero identified curvature is consistent with an affine path (Theorem 1(b)), so identified curvature can refute

nonlinear shapes of conditioning but cannot by itself establish that conditioning has stopped; see Proposition 1.

Proof. The first sentence is Theorem 1(a). Under C_d , $\mathcal{K}_\tau$ is spanned by $s - 1$ and the residue-class indicators $\mathbf{1}\{s \equiv r\}$, $r \neq 1$ (Theorem 1(c)), and orthogonality to that basis is the displayed pair. The weights of $\Delta_d^2 \tau(s)$ are $(1, -2, 1)$ at $s, s + d, s + 2d$, which lie in one residue class and sum to zero there, and $\sum \lambda_s(s - 1) = (s + 2d - 1) - 2(s + d - 1) + (s - 1) = 0$. For $d > 1$ the weights of $\Delta^2 \tau(s)$ occupy three distinct residue classes with nonzero sums, and those of $\tau(s + d) - \tau(s)$ have $\sum \lambda_s(s - 1) = d \neq 0$. $\square$

The stylized fact that “conditioning happens in the first few waves” is thus normalization-free only in designs with adjacent entry cohorts that satisfy C_1 ; in a panel refreshed at a stride $d > 1$, the shape of conditioning is identified only on the coarser grid of that stride, and every published *level* is a normalization in every design. This is the sense in which a refreshment schedule is an identification decision.

3.1 Interrupted participation: rotation designs

M1 assumes uninterrupted participation, so that the number of interviews a unit has received coincides with its calendar tenure. Rotation designs violate it by construction. The U.S. Current Population Survey interviews a household for four consecutive months, rests it for eight, and returns it for four more—the 4–8–4 pattern whose rotation-group differences are the subject of the literature this paper began with (Bailar 1975; Solon 1986; Krueger, Mas, and Niu 2017; Halpern-Manners and Warren 2012)—so that its eighth interview arrives in its sixteenth month. The relaxation is worth making because the two clocks it separates are exactly the two sides of Theorem 1: conditioning is a function of the *dose*, the number of interviews received, while the linear dependency that makes the path unidentified is a statement about *calendar* time. When the two coincide, the dependency is unbreakable. When they do not, it is.

Let the entry cohorts be $\mathcal{E}$ with stride d as before, and let every cohort follow a common *interview pattern* $J = \{j_1 = 0 < j_2 < \dots < j_K\}$ of offsets from entry: cohort e is interviewed

at the calendar times $e + j_k$ that lie in the field period, and the interview at $e + j_k$ is its k -th, so its dose is k . Write $j(k) := j_k$ for the elapsed calendar time at which the k -th interview arrives. The cell means are

$$\mu(e, e + j_k) = \alpha(e + j_k) + g(e) + \tau(k), \quad \tau(1) = 0, \quad g(e_0) = 0, \quad (2)$$

which is M4 with the dose in place of the tenure; uninterrupted participation is the case $J = \{0, 1, \dots, K - 1\}$, where $j(k) = k - 1$ and Equation 2 is Equation 1. We assume:

- **(P) Pattern connectivity.** Every spacing in $d\mathbb{Z} \cap [1, j_K]$ is realized by two cohorts of the sample that are observed at a common period. For continuous recruitment at stride d this holds as soon as the entry window is at least as long as the pattern's span.

Theorem 2 (Interrupted participation: the calendar clock is the one that cannot be seen).

Under M2–M3, Equation 2, and a common interview pattern J :

- (a) (Directions that are never identified.) *For every $m \in \mathbb{R}$ and every d -periodic ρ with $\rho(0) = 0$, the vector $h_\tau(k) = m j(k) + \rho(j(k))$, $h_g(e) = m(e - e_0)$, $h_\alpha(t) = -m(t - e_0) - \rho(t - e_0)$ lies in the kernel of the cell design.*
- (b) (Sharpness.) *Under (P), and on a connected cohort–period incidence graph, these exhaust it:*

$$\Theta_\tau = \{ k \mapsto \tau(k) + m j(k) + \rho(j(k)) : m \in \mathbb{R}, \rho \text{ } d\text{-periodic}, \rho(0) = 0 \},$$

whose dimension is the number of residue classes modulo d represented in J —one when $d = 1$.

- (c) (Identified functionals.) *$\lambda' \tau$ is identified if and only if $\sum_k \lambda_k j(k) = 0$ and, for $d > 1$, $\sum_{k: j(k) \equiv r} \lambda_k = 0$ for every residue $r \not\equiv 0$ represented in J . Equivalently: writing $T(j) := \tau(j)$ for the path as a function of elapsed calendar time, what is identified is exactly the second-order divided differences of T on J —the curvature of conditioning against the calendar, not against the dose.*
- (d) (When the dose is separable from the calendar.) *A path that is linear in the dose, $\tau(k) =$*

$c(k-1)$, belongs to the identified set of the null path—and is therefore indistinguishable from no conditioning at all—if and only if $k-1 = mj(k) + \rho(j(k))$ for some such (m, ρ) ; with $d = 1$ this says precisely that the interviews are equally spaced. A design whose interviews are unequally spaced can distinguish conditioning that accumulates per interview from conditioning that accumulates per unit of calendar time; a design whose interviews are equally spaced—including every uninterrupted design—cannot distinguish either from nothing.

Proof. (a) At the cell $(e, t) = (e, e+j_k)$ we have $j(k) = t-e$, so $mj(k) = m(t-e_0) - m(e-e_0)$ cancels against h_α and h_g ; and $\rho(j(k)) = \rho(t-e) = \rho(t-e_0)$ because d divides $e-e_0$, which cancels against the second term of h_α . The normalizations hold since $\rho(0) = 0$.

(b) Let h be in the kernel and put $H(j) := h_\tau(k(j))$ for $j \in J$, so $H(0) = h_\tau(1) = 0$. If cohorts e and $e-\delta$ are both observed at a period t , their offsets are $j = t-e$ and $j+\delta$, both in J , and subtracting the two cell equations gives

$$h_g(e) - h_g(e-\delta) = H(j+\delta) - H(j).$$

The left-hand side does not depend on t , so for each realized spacing δ the increment $G(\delta) := H(j+\delta) - H(j)$ is the same at every j for which both offsets lie in J ; and G is additive, $G(\delta+\delta') = G(\delta) + G(\delta')$, by telescoping h_g across cohorts. An additive function on $d\mathbb{Z}$ is linear, so $G(\delta) = m\delta$ with $m := G(d)/d$. Now take $j, j' \in J$ in the same residue class modulo d ; their difference is a spacing in $d\mathbb{Z} \cap [1, j_K]$, realized by (P), so $H(j) - H(j') = m(j-j')$: on each residue class $H(j) - mj$ is constant. Writing ρ for that constant as a function of the class— $\rho(0) = 0$ by $H(0) = 0$ —gives $H(j) = mj + \rho(j)$, which is (a); and the free parameters are m together with one constant per represented class other than that of 0. Back-substituting, $h_\alpha(t) + h_g(e) = -m(t-e) - \rho(t-e_0)$ splits as in the proof of Theorem 1(c), uniquely on each component of the incidence graph and hence globally when it is connected.

(c) is Theorem 1(a) with the kernel of (b): the functional is constant on Θ_τ iff it annihilates $j(\cdot)$ and each class indicator. The divided-difference statement is the observation that the functionals annihilating $\{1, j\}$ on a finite set J are spanned by the second-order divided

differences of T .

(d) is (b) read as a membership question, together with the fact that, for $d = 1$, $k-1 = m j(k)$ for all k holds iff $j(k)$ is a constant multiple of $k - 1$, i.e. iff the offsets are equally spaced.

□

□

Corollary 2 (What a 4–8–4 rotation identifies). *Take the CPS pattern $J = \{0, 1, 2, 3, 12, 13, 14, 15\}$ with monthly entry ($d = 1$) and an entry window long enough for (P)—at least ten cohorts, the largest gap in the pattern being nine months. Then Θ_τ is one-dimensional, $\{\tau + m j(\cdot)\}$, and:*

(i) *no level $\tau(k)$, $k \geq 2$, and no increment $\tau(k + 1) - \tau(k)$ is identified;*

(ii) *every difference of per-month rates is identified: for any two interview intervals,*

$$\frac{\tau(k + 1) - \tau(k)}{j(k + 1) - j(k)} - \frac{\tau(l + 1) - \tau(l)}{j(l + 1) - j(l)}$$

is a point-identified functional. In particular

$$\mathcal{D} := [\tau(5) - \tau(4)] - 9[\tau(2) - \tau(1)]$$

is identified, and it is computable from published month-in-sample means;

(iii) *$\mathcal{D} = 0$ for every path that accumulates at a constant rate per calendar month, and $\mathcal{D} = -8c$ for a path that accumulates at a constant rate c per interview. The 4–8–4 design therefore answers, from cell means alone, a question no uninterrupted design can pose: whether conditioning is a function of interviews taken or of time enrolled.*

Proof.

(i) and (ii) are Theorem 2(c): the weights of a level or of a single increment do not annihilate $j(\cdot)$, while those of a difference of per-month rates do, since each rate contributes $\{j(k+1) - j(k)\} / \{j(k+1) - j(k)\} = 1$. For $\mathcal{D}$, $\sum_k \lambda_k j(k) = (12 - 3) - 9(1 - 0) = 0$. (iii) Substitute $\tau(k) = a + b j(k)$, which gives $b(12 - 3) - 9b(1 - 0) = 0$; and $\tau(k) = c(k - 1)$, which gives $c(4 - 3) - 9c(1 - 0) = -8c$. □

□

Three comments. First, everything downstream is unchanged: Theorem 3 is stated for an arbitrary element of $\ker X_{\mathcal{G}}$, so two-way fixed effects annihilate the unidentified component of a dose design exactly as they do that of a tenure design, and the corrected estimator of Theorem 4 is the same regression with *dose* indicators in place of tenure indicators. What changes is which component is unidentified— $m j(k)$ rather than $m(k - 1)$ —and therefore which functionals the correction can use. Second, the mechanism is the mirror image of the one in the Antecedents paragraph above: unequal spacing between *entry cohorts* creates periodic directions that no schedule can see, while unequal spacing between *a unit's own interviews* destroys an affine direction that every equally spaced schedule must live with. Identification in this model is entirely a question about which functions of elapsed calendar time can be written as a function of the period plus a function of the cohort. Third, (P) is a real condition and not a formality: with the CPS pattern and fewer than ten entry cohorts the sample cannot bridge the eight-month rest, and the identified set is two-dimensional rather than one (verified in the replication package, which also reproduces every row of the table below).

Table 1: What a panel's schedule decides before any data are collected. The dimension is that of the identified set of the conditioning path; the last column asks whether the accumulation clock is visible—whether a path accumulating at a constant rate per interview can be distinguished from no conditioning at all. Every row is verified by exact rank computation in the replication package.

			dose
Schedule: entry stride; interview			clock
pattern	$\dim \Theta_{\tau}$	identified functionals of τ	visible?
adjacent entry; uninterrupted	1	all second differences $\Delta^2\tau$	no
entry every d waves; uninterrupted	d	lag- d second differences only	no
JLPS: entries at waves 1, 5, 13; uninterrupted; $T = 19$	4	$\Delta_4^2\tau$ only	no
entries at 1, 4, 8; uninterrupted; $T = 8$ (no follow-up)	3	only within increment-graph components	no

			dose
Schedule: entry stride; interview			clock
pattern	$\dim \Theta_\tau$	identified functionals of τ	visible?
single refreshment at wave e_2 ; uninterrupted	$e_2 - 1$	$\Delta_{e_2-1}^2 \tau$ only	no
monthly entry; CPS 4–8–4 pattern	1	differences of per-month rates (Corollary 2)	yes
adjacent entry; 2 on, 2 off, 2 on	1	differences of per-month rates	yes
adjacent entry; every other period (1 on, 1 off)	1	second differences in the dose	no
entry every 2 periods; CPS 4–8–4 pattern	2	the above, plus a parity restriction	yes

The table is the practical residue of this section. Two decisions fix what a panel can ever learn about conditioning: how far apart its entry cohorts are, which sets the dimension of what cannot be known, and how far apart a unit’s own interviews are, which decides whether the accumulation clock is visible at all. Neither decision is usually taken on identification grounds; both are cheap to compute before fielding, and Theorem 2 says the second one is free—a rest period that a rotation design has for cost reasons turns out to be the only reason its month-in-sample differences can be read as a dose response.

4 Attrition and Refreshment Samples: What Changes

Theorem 1 assumes M3, no attrition. Relaxing it changes the problem in a way that this paper does not solve but must locate. With nonignorable attrition, the stayers’ cell means are contaminated by selection on the latent outcome, and a refreshment sample—a fresh cross-section drawn at a later wave—identifies the population marginal that the retained sample no longer represents. The companion paper (Okubo 2026, Theorem 1) characterizes exactly what such a sample identifies about a conditioning map when attrition is unrestricted: the

retention-scaled distribution of the stayers’ implied latent outcomes must be dominated, set by set, by the refreshment distribution, and every map satisfying this constraint is rationalized by an explicitly constructed admissible selection process. Three consequences of that characterization matter here. First, when attrition is independent of the latent outcome, a location shift in the reports is point identified from the refreshment sample even at interior retention whenever the refreshment density has tails thinner than exponential (Okubo 2026, Corollary 3): attrition that does not select on the outcome cannot masquerade as conditioning. Second, when attrition does select on the outcome, the identified *set* for a location shift is governed by how fast the selection probability can decay in the tails of the refreshment distribution; it is an interval when the refreshment distribution has a unimodal density (Okubo 2026, Theorem 3), and it need not be connected otherwise—with a refreshment distribution placing mass .5 on each of two points, retention .4, and stayers all reporting the lower point, the feasible latent locations are the two points and nothing between them. Third, the panel’s own longitudinal structure does not narrow that set (Okubo 2026, Theorem 2); separation is restored only by additional information—negative-control items measured once at entry, or restrictions linking selection across waves. This paper does not use any Gaussian construction of selection processes; the companion paper’s completion argument is the treatment we rely on.

5 Recovery: Identifying Restrictions

Each result below adds a restriction that removes part or all of $\mathcal{K}_\tau$, and each is stated together with what is, and is not, testable about it. Throughout this section C_d is assumed, so that $\mathcal{K}_\tau$ is the affine-plus-periodic family of Theorem 1(c); on a support violating C_d the same restrictions are checked against the computed $\mathcal{K}_\tau$.

Proposition 1 (Plateau (saturation) as an identifying restriction). *Suppose $\tau(s) = \bar{\tau}$ for all $s \geq S^*$, and suppose the observed tenure range contains at least $d + 1$ consecutive tenures $s \geq S^*$. Then Θ_τ collapses to a point: the restriction forces $m = 0$ and $\rho \equiv 0$ in Theorem 1(c), so τ is point identified on the observed tenure set. The restriction has testable observable implications: every identified functional $\lambda'\tau$ of Corollary 1 that is supported on tenures*

$s \geq S^*$ and has zero-sum weights, $\sum_s \lambda_s = 0$, must vanish—in particular $\Delta_d^2 \tau(s) = 0$ for $s \geq S^*$ and, with three or more cohorts, the constancy over time of the cohort gaps observed within the plateau window. Identified functionals whose weights do not sum to zero, such as $2\tau(5) - \tau(9)$ on a stride-four support, are not implied to vanish—on a plateau they equal $\bar{\tau} \sum_s \lambda_s$. The restriction itself is not testable: the affine path $\tau(s) = m(s - 1)$, $m \neq 0$, satisfies every observable implication and does not saturate. A non-rejection therefore supports the plateau only against nonlinear departures; the zero-slope part of the restriction must be argued on substantive grounds or supplied by an external anchor (Proposition 2, or a refreshment design of the kind treated in the companion paper). The restriction is the measurement-domain form of the oldest identifying device in the age–period–cohort literature: Mason et al. (1973) break the linear dependency by assuming that two categories within one dimension have identical effects—“under the assumption that any two ages, periods or cohorts have identical effect parameters, differences of the form ... are now estimable”—and a plateau is that assumption imposed on a block of adjacent tenures rather than on a pair.

Proof. On the plateau range $h_\tau(s) = m(s - 1) + \rho(s)$ must be constant. The range contains some s and $s + d$, and $\rho(s + d) = \rho(s)$ gives $md = 0$, so $m = 0$; then ρ is constant on $d + 1$ consecutive tenures, hence on every residue class, hence everywhere, and it equals $\rho(1) = 0$. For the observable implications, an identified functional supported on the plateau takes the value $\bar{\tau} \sum_s \lambda_s$ there, which is zero iff the weights sum to zero. For untestability, $\tau(s) = m(s - 1)$ lies in Θ_τ of the null path and has zero identified curvature everywhere. $\square$ $\square$

Proposition 2 (Entry-wave negative controls). *Let N be a set of items measuring time-invariant facts, each reported exactly once per respondent at entry ($s = 1$). By construction $\tau_N \equiv 0$, so entry contrasts across cohorts identify $g_N(e) - g_N(e')$ plus sampling and selection differences. Suppose that (i) recall of the fact is stable across the calendar times at which different cohorts enter, (ii) the instrument and its administration are the same for every cohort, (iii) the entry contrasts are computed on the full entry cohorts, before any attrition (M3), or after a correction for selection of the kind the companion paper supplies, and (iv) cohort effects transport to the target item j with a known or bounded loading, $g_j(e) = \Gamma_j g_N(e)$.*

Then the negative controls identify (or bound) the cohort effects g_j , and knowledge of g_j removes the affine direction of $\mathcal{K}_\tau$: the coefficient m in Theorem 1(c) is pinned, because $h_g(e) = m(e - e_0)$ must vanish. It does not remove the periodic directions, which leave g unchanged. Consequently: if $d = 1$, τ_j is point identified (or bounded by the range of Γ_j); if $d > 1$, the negative controls identify $\tau_j(s)$ at the tenures $s \equiv 1 \pmod{d}$ and the identified functionals of Corollary 1 combined with such levels, while $\tau_j(s)$ at other tenures remains identified only up to the $d - 1$ periodic directions, whatever the loading. Point identification of the full path with $d > 1$ requires, in addition, a restriction that removes the periodic directions—the plateau of Proposition 1 spanning $d + 1$ consecutive tenures, or an external anchor of the periodic component. Conditions (i)–(iv) are assumptions about the items and the sample, not consequences of measuring the same respondents: an immutable fact does not imply a stable recall, and a common battery does not imply a common loading. We therefore recommend reporting the sensitivity of the recovered path to Γ_j over a substantive range and, where no range can be defended, treating the battery as a diagnostic of cohort comparability rather than as an identifying instrument. Absent any restriction linking g_j across cohorts, Theorem 1 applies item by item.

Proof. Under (i)–(iv), g_j is known (or bounded) on $\mathcal{E}$. Two triples generating the same cell means differ by $h \in \mathcal{K}_0$ with $h_g(e) = m(e - e_0)$; if both have the known g_j , then $m(e - e_0) = 0$ at some $e \neq e_0$, so $m = 0$, and the remaining freedom is $h_\tau = \rho$, d -periodic with $\rho(1) = 0$, which vanishes at $s \equiv 1 \pmod{d}$ and is identically zero iff $d = 1$. On the stride-four support of Section 8, for instance, $\rho(s) = a \mathbf{1}\{s \equiv 2 \pmod{4}\}$ with $h_\alpha(t) = -a \mathbf{1}\{t \equiv 2 \pmod{4}\}$ and $h_g \equiv 0$ preserves every cell mean, every entry-wave negative-control contrast, and the exact value of g , while moving $\tau(2)$ by an arbitrary a . $\square$

Proposition 3 (Partial identification under bounded cohort drift). *Suppose cohort effects drift at a bounded per-period rate: for all consecutive entry cohorts $e_k < e_{k+1}$,*

$$\left| \frac{g(e_{k+1}) - g(e_k)}{e_{k+1} - e_k} \right| \leq M.$$

The periodic directions of Theorem 1 do not move g , so this restriction constrains only

the affine coefficient m . Fix any representative $(\hat{\alpha}, \hat{g}, \hat{\tau}) \in \Theta$ and let $\hat{d}_k = \{\hat{g}(e_{k+1}) - \hat{g}(e_k)\}/(e_{k+1} - e_k)$ be its implied drifts. The true triple has $g = \hat{g} + m(e - e_0)$ for some m , so $|\hat{d}_k + m| \leq M$ for every k , giving $m \in [-M - \min_k \hat{d}_k, M - \max_k \hat{d}_k]$. Consequently, for every tenure $s \equiv 1 \pmod{d}$ —the tenures at which the periodic component vanishes—the bounds

$$\tau(s) - \tau(1) \in [\hat{\tau}(s) + m_{lo}(s - 1), \hat{\tau}(s) + m_{hi}(s - 1)]$$

are sharp, with endpoints linear in M ; the same holds for any identified functional of Corollary 1 combined with such a level. At tenures $s \not\equiv 1 \pmod{d}$ the periodic component remains free and the identified set for $\tau(s) - \tau(1)$ is unbounded for every M . The interval is nonempty iff $\max_k \hat{d}_k - \min_k \hat{d}_k \leq 2M$, so with three or more cohorts the bound is refutable; $M = 0$ delivers point identification at the covered tenures together with an overidentification test ($\hat{d}_1 = \hat{d}_2$); $M \rightarrow \infty$ recovers Θ_τ . Reporting the breakdown value M^* at which conclusions reverse is the recommended default output. In a panel refreshed at stride d the covered tenures are $1, 1 + d, 1 + 2d, \dots$ —in the Japanese panel of Section 8, tenures 5, 9, 13, and 17, which are exactly the tenures at which its refreshment samples enter and at which its increments are estimated.

Why first differences and not second. The affine direction $g(e) + m(e - e_0)$ is linear in e , so any restriction on second differences of g —smoothness in the Rambachan–Roth sense—is invariant along Θ_τ and yields no information about m : under $|\Delta^2 g| \leq M$ the identified set for $\tau(s) - \tau(1)$ remains the whole real line for every M . This is the measurement-domain image of the APC fact that it is only the slopes of an APC model that are unidentified, so that constraints invariant to the linear component cannot recover it, and bounds must instead be placed on the slopes themselves (Fosse and Winship 2019b).

6 Where the Unidentified Component Goes: Downstream Propagation

Theorem 1 says what a panel cannot reveal about conditioning. This section asks what that ignorance costs the analyst who uses the panel for something else—a fixed-effects regression, a

difference-in-differences, an event study—and finds that the answer is exactly complementary: the component of τ that cannot be identified is the component that the workhorse panel estimators discard, and the component they do not discard is identified. Identification failure upstream and robustness downstream are the same piece of linear algebra.

6.1 Setting

Keep the tenure structure of Section 2: individual i enters at e_i and is observed at calendar times t with tenure $s_{it} = t - e_i + 1$. An analyst is interested in an outcome Y_{it}° —the response the individual would give at tenure one, $Y_{it}^\circ := Y_{it}(1)$, the unconditioned response—and possibly in the effect of a regressor D_{it} on it. The analyst observes the report

$$Y_{it} = Y_{it}^\circ + \tau(s_{it}) + \nu_{it}, \quad (3)$$

where τ is the conditioning path of Equation 1 and ν_{it} collects individual departures from the mean path. We maintain:

- **D1 (mean-additive drift).** $\mathbb{E}[\nu_{it} \mid \{D_{i't'}, s_{i't'}\}_{i',t'}] = 0$: the conditioning effect may be heterogeneous, but its mean given the design is the common path $\tau(s)$. Heterogeneity that is itself correlated with D is excluded by D1; it is a separate, standard endogeneity problem and is not the subject here. Every statement below is a statement about a common additive tenure effect satisfying D1.
- **D2 (uninterrupted participation).** Units are interviewed at every wave from entry until they leave, so that calendar tenure $t - e_i + 1$ equals the number of interviews received; interrupted rotation designs such as the CPS 4–8–4 pattern, in which the two differ, are not treated.

Let W be the $n \times (N + T)$ matrix of unit and calendar-time indicators for the observed rows (any unbalanced support), $M = I - W(W'W)^{-1}W'$ the residual-maker that projects off unit and time effects, and for any n -vector v write $v(s)$ for the vector with entries $v(s_{it})$. Throughout, “TWFE” means the OLS coefficient on D in a regression with unit and time effects, $\hat{\beta} = (D'MD)^{-1}D'MY$, on whatever support is observed. The two-way

within transformation on a balanced panel is the special case $MD = D_{it} - \bar{D}_i - \bar{D}_t + \bar{D}$; nothing below requires balance.

6.2 The absorption theorem

Theorem 3 (Absorption of the unidentified component by two-way fixed effects). *Under D1, let $\hat{\beta}$ be the TWFE coefficient computed on the reports Y and $\hat{\beta}^\circ$ the (infeasible) coefficient computed on the unconditioned outcomes Y° , both on the same support with the same regressor. Then, conditional on the design (D, s) ,*

$$\mathbb{E}[\hat{\beta} - \hat{\beta}^\circ] = (D'MD)^{-1}D'M\tau(s) = \frac{\sum_{it} \ddot{D}_{it} \tau(s_{it})}{\sum_{it} \ddot{D}_{it}^2}, \quad \ddot{D} := MD, \quad (4)$$

and the following hold.

- (i) Absorption. *Let $\mathcal{S}$ be the cohort–period support of the estimation sample and Z_τ the matrix of tenure indicators $\mathbf{1}\{s_{it} = s\}$. For every $h = (h_\alpha, h_g, h_\tau) \in \ker X_\mathcal{S}$, $MZ_\tau h_\tau = 0$. In particular $M\{a + bs + \rho(s)\} = 0$ for every $a, b \in \mathbb{R}$ and every d -periodic ρ , d the stride of the entry cohorts in the sample. Hence the bias in Equation 4 is unchanged when τ is replaced by any other member of the identified set Θ_τ of Theorem 1(a) computed on the same support: it is invariant along the entire identified set, on any support, whether or not C_d holds.*
- (ii) Exactness for unidentified drift. *If $\tau \in \mathcal{K}_\tau$ —the identified set of the null path on the sample’s support; under C_d , an affine function of s plus a d -periodic function—then $\mathbb{E}[\hat{\beta} - \hat{\beta}^\circ] = 0$.*
- (iii) Only the identified part matters. *Writing $\tau = \tau_\parallel + \tau_\perp$ for any decomposition into a member of $\mathcal{K}_\tau$ and a remainder, the bias equals $(D'MD)^{-1}D'M\tau_\perp(s)$; in particular it is zero if and only if $\ddot{D} \perp \tau_\perp(s)$.*

Proof. By the Frisch–Waugh–Lovell theorem, $\hat{\beta} = (D'MD)^{-1}D'MY$ and $\hat{\beta}^\circ = (D'MD)^{-1}D'MY^\circ$ on the same support, so by Equation 3, $\hat{\beta} - \hat{\beta}^\circ = (D'MD)^{-1}D'M\{\tau(s) + \nu\}$, and D1 removes the second term in expectation. Since M is symmetric and idempotent,

$D'M\tau(s) = (MD)'\tau(s) = \sum_{it} \ddot{D}_{it}\tau(s_{it})$. For (i), let $h \in \ker X_S$. The cell equation $h_\alpha(t) + h_g(e) + h_\tau(t - e + 1) = 0$ holds for every fielded cell, so for every observed row, $(Z_\tau h_\tau)_{it} = h_\tau(s_{it}) = -h_\alpha(t) - h_g(e_i)$: a function of calendar time plus a function of the unit (through its entry cohort). Both lie in $\text{col}(W)$, so $Z_\tau h_\tau \in \text{col}(W)$ and $MZ_\tau h_\tau = 0$. The affine and periodic directions are the elements of $\mathcal{K}_0 \subseteq \ker X_S$ of Theorem 1(b), which gives the special case; directly, $s_{it} = t + (1 - e_i)$ and $\rho(s_{it}) = \rho(t - e_0 + 1)$ when all cohorts are congruent modulo d . Replacing τ by another member of Θ_τ adds $D'MZ_\tau h_\tau = 0$ to the numerator. (ii) is the case $\tau \in \mathcal{K}_\tau$. (iii) follows from (i) by linearity: $D'M\tau(s) = D'M\tau_{\parallel}(s) + D'M\tau_{\perp}(s) = D'M\tau_{\perp}(s)$, and the bias is zero iff $\ddot{D}'\tau_{\perp}(s) = 0$. $\square$

Three remarks. First, Theorem 3 is an exact finite-sample statement conditional on the design, not a large- N approximation, and it holds on any unbalanced support: attrition, rotation, and refreshment do not disturb the argument because they change the rows of W , not the fact that s_{it} is a sum of a unit term and a time term. Second, the theorem does not say that TWFE is immune to conditioning; it says that TWFE is immune to *exactly the part of conditioning that no one can measure*, and exposed only to the identified part of Corollary 1. The periodic directions, which are not affine, do not bias fixed-effects estimates either: on any sample whose cohorts share a stride they are functions of calendar time, and on any sample at all every kernel direction's tenure component is a sum of a time function and a unit function, which is the one-line content of part (i). Third, the same argument covers any estimator that partials out unit and time effects before using D —the two-way Mundlak, the interactive-fixed-effects estimators with unit and time effects included additively, and first differences with time effects, since $\Delta\tau(s_{it})$ is constant for affine τ and absorbed by the intercept.

Corollary 3 (Protected and exposed designs).

- (a) Single-cohort panels. *If all units in the estimation sample share one entry cohort, then s_{it} is a function of t alone, $M\tau(s) = 0$ for **every** path τ , and every TWFE, difference-in-differences, and first-difference coefficient computed within the cohort has zero measurement-induced bias relative to its unconditioned counterpart, under tenure*

drift of arbitrary shape. This is a statement about the incremental bias due to conditioning, not about the causal validity of the unconditioned estimand.

- (b) Tenure-orthogonal treatment. More generally, the bias vanishes for every τ if and only if the double-demeaned regressor sums to zero within every tenure level, $\sum_{it: s_{it}=s} \ddot{D}_{it} = 0$ for all s in the support. When treatment timing is assigned independently of entry cohort given calendar time, $\mathbb{E}[D_{it}]$ is a function of t alone, so $\mathbb{E}[\ddot{D}] = M \mathbb{E}[D] = 0$ and the numerator of Equation 4 has mean zero over the assignment distribution.
- (c) Level contrasts. Estimands that compare levels across units of different tenure at a common period—pooled cross-section comparisons, cohort contrasts, panel-versus-fresh-sample benchmarks, and the “experienced versus new respondent” contrasts of the conditioning literature itself—identify the sum $g(e) - g(e') + \tau(s) - \tau(s')$, which is invariant along Θ_τ ; but its decomposition into a cohort difference and a conditioning difference is not identified, and the conditioning component $\tau(s) - \tau(s')$ that such contrasts are meant to measure inherits the non-identification of Theorem 1 unless a normalization or recovery condition of Section 5 is invoked.

Proof.

- (a) With a common e , $\tau(s_{it}) = \tau(t - e + 1)$ depends on (i, t) only through t , so $\tau(s) \in \text{col}(W)$ and $M\tau(s) = 0$. (b) The numerator of Equation 4 is $\sum_{it} \ddot{D}_{it} \tau(s_{it}) = \sum_s \tau(s) \sum_{it: s_{it}=s} \ddot{D}_{it}$, which is zero for every τ if the stated condition holds, and which is nonzero for $\tau = \mathbf{1}\{s = s_0\}$ whenever the condition fails at s_0 . For the second claim, $\ddot{D} = MD$ is linear in D , so $\mathbb{E}[\ddot{D}] = M \mathbb{E}[D]$, and a function of t alone lies in $\text{col}(W)$. (c) is Equation 1 directly: a level contrast at a common t between cohorts e, e' with tenures $s = t - e + 1$, $s' = t - e' + 1$ equals $g(e) - g(e') + \tau(s) - \tau(s')$. Along $\mathcal{K}_0$ the conditioning component moves by $m(s - s') + \rho(s) - \rho(s')$ and the cohort component by $m(e - e')$; since $s - s' = e' - e$ and $\rho(s) = \rho(s') = \rho(t - e_0 + 1)$, the two movements cancel and the sum is unchanged, while each component moves. $\square$

$\square$

Part (a) is the reassurance the applied literature has been implicitly relying on: a panel that recruited everyone at once cannot have its within-unit estimates contaminated by a common additive tenure effect satisfying D1, whatever its shape. Part (c) is the warning: the moment a design mixes tenures within a period—every rotating panel, every panel with refreshment samples, every pooled analysis of an old and a new cohort—the observed level contrast is identified but its conditioning component is not, so its reading as a conditioning effect is normalization-dependent; the TWFE protection of Theorem 3, by contrast, extends to the whole unidentified component, and what it does not cover is the identified curvature.

6.3 The bias is identified even though the level is not

Theorem 1 and Theorem 3 combine into a constructive statement that we regard as the practical center of the paper.

Theorem 4 (Identified correction of two-way fixed-effects estimates). *Let $\lambda := MD(D'MD)^{-1}$ be the implicit weights of the TWFE estimator, so that the bias in Equation 4 is the linear functional $b(\tau) = \sum_{it} \lambda_{it} \tau(s_{it}) = \sum_s \Lambda_s \tau(s)$ with $\Lambda_s = \sum_{it: s_{it}=s} \lambda_{it}$. Then $\Lambda \perp \mathcal{K}_\tau$ for the kernel of the estimation sample's own cell design—under C_d , the pair of conditions of Corollary 1, $\sum_s \Lambda_s(s-1) = 0$ and $\sum_{s \equiv r} \Lambda_s = 0$ for every residue class r modulo the stride of the sample. Consequently $b(\tau)$ is point identified whenever τ is identified up to Θ_τ on the estimation sample (M1–M4 on that sample, or an external estimate of the shape of τ on a design whose kernel is no larger). A feasible corrected estimator is the coefficient on D in the joint regression*

$$Y = W\gamma + \beta D + Z_\tau \theta + \text{error}, \quad (5)$$

where Z_τ is the matrix of tenure indicators $\mathbf{1}\{s_{it} = s\}$ and the columns of Z_τ that are collinear with W —exactly the directions of Θ_τ —are dropped. Suppose that, on the true outcome, $\mathbb{E}[Y^\circ \mid D, W, s] = W\gamma + \beta D$, and that

- **(R) Residual rank.** *After partialling out W , the regressor D is not collinear with the retained tenure indicators: $[MD, MZ_\tau]$ has full column rank.*

Then, under D1–D2, the joint OLS coefficient $\hat{\beta}^c$ satisfies $\mathbb{E}[\hat{\beta}^c] = \beta$ exactly, conditional on the design; equivalently $\hat{\beta}^c = \hat{\beta} - (D'MD)^{-1}D'MZ_\tau\hat{\theta}$ for any representative $\hat{\theta}$ of the identified set, so the correction subtracts an estimate of $b(\tau)$ that does not depend on which representative is used. No normalization of the conditioning level is required. Condition (R) fails, and β is not identified, when treatment status is a deterministic function of tenure on the sample (for instance $D_{it} = \mathbf{1}\{s_{it} \geq s^*\}$), or more generally when MD lies in the span of the retained tenure indicators; in that case the joint regression returns no coefficient on D rather than a wrong one.

Proof. For any $h \in \ker X_s$, $\sum_s \Lambda_s h_\tau(s) = \lambda' Z_\tau h_\tau = (D'MD)^{-1}D'MZ_\tau h_\tau = 0$ by Theorem 3(i), so $\Lambda \perp \mathcal{K}_\tau$ and, by Theorem 1(a), b is constant on Θ_τ ; under C_d the same follows from $\Lambda'1 = 0$, $\Lambda'(s-1) = 0$ (because $1, s \in \text{col}(W)$) and $\sum_{s \equiv r} \Lambda_s = \lambda' \mathbf{1}\{s \equiv r\} = 0$ (the residue-class indicator is a function of t on the sample). For the feasible estimator, the reported outcome satisfies $Y = Y^\circ + Z_\tau \tau + \nu$ exactly, since $\tau(s_{it}) = (Z_\tau \tau)_{it}$; hence $Y = W\gamma + \beta D + Z_\tau \tau + \{Y^\circ - W\gamma - \beta D\} + \nu$, and the bracketed term and ν have conditional mean zero by the outcome model and D1. Dropping the columns of Z_τ collinear with W changes θ to another representative of Θ_τ and leaves the fitted values, hence the coefficient on D , unchanged. Under (R) the remaining design $[W, D, Z_\tau]$ has full column rank up to $\text{col}(W)$, so OLS is unbiased for β ; the equivalent form follows from the Frisch–Waugh–Lovell theorem applied to D and Z_τ jointly. If $MD \in \text{col}(MZ_\tau)$ the coefficient on D is not determined and OLS reports it as aliased. $\square$

The result inverts the usual reading of an impossibility theorem. The level of conditioning cannot be known; but the quantity the applied analyst needs—the contamination of her fixed-effects coefficient—does not depend on the level, and is recoverable from the panel’s own staggered structure or from an external estimate of the conditioning *shape* (a refreshment design, or the shape library assembled in companion work). Three cautions accompany it. The correction removes the implied measurement component when the outcome model and D1 hold; its uncertainty includes the estimation of that component, and the joint regression’s variance estimator must respect the dependence in the data—we use standard errors clustered by unit—whereas a two-step procedure that first calibrates τ from the same outcome’s cell

means and then subtracts it neither propagates that uncertainty nor, if the cell-mean model omits D , keeps the genuine effect out of the estimated tenure component. Condition (R) is a real restriction: designs in which exposure is scheduled by tenure cannot separate the effect of exposure from conditioning at all. And the correction is a *within-sample* construction: it uses the estimation sample’s own staggered structure to estimate the identified part of τ , and it is not available when that structure is absent, as in the exactly aligned event studies of Section 6.4, where an *external* estimate of τ must be supplied instead. Simulation 4 (Appendix C) verifies the identity, reports the finite-sample behavior of Equation 5 under five designs including one that violates (R), and shows the practical caveat that with three entry cohorts and stride four the identified curvature rests on few degrees of freedom, so the correction is exact in expectation and imprecise in practice; the plateau restriction of Proposition 1 or an external shape estimate are the routes to precision.

6.4 Spurious pre-trends in tenure-aligned event studies

The protection of Theorem 3 has a precise boundary, and event studies sit on it. Let E_i be the calendar time of an event for unit i and $k_{it} = t - E_i$ event time. The fully dynamic event-study specification regresses the outcome on unit effects, time effects, and indicators $X_k = \mathbf{1}\{k_{it} = k\}$ for event times k in a window $\mathcal{K}$, omitting a set of reference categories. It is well known (Borusyak, Jaravel, and Spiess 2024; Sun and Abraham 2021) that when every unit is eventually treated a complete set of event-time indicators is collinear with unit and time effects in two directions—a constant, and a linear trend in k , since $k_{it} = t - E_i$ is the sum of a time term and a unit term—so that two reference categories must be omitted (or an untreated group included) for the coefficients to be defined. The second of those directions is the same one as in Theorem 1, and it is exactly what makes tenure drift masquerade as dynamics.

Theorem 5 (Mechanical pre-trends under exact tenure–event alignment). *Suppose the design aligns tenure with event time: $s_{it} = k_{it} + \delta$ for a constant δ on the estimation sample (equivalently $E_i = e_i + \delta - 1$: every unit’s event falls a fixed number of waves after its entry, as in rotation panels, event-recruited samples, and surveys fielded on entry schedules). Let*

the analyst estimate the fully dynamic specification with unit and time effects and indicators X for $k \in \mathcal{K} \setminus \{k_0, k_1\}$, $k_0 \neq k_1$ two reference categories, and assume that the event-time indicators are not collinear with the fixed effects: MX has full column rank, where M projects off unit and time effects (equivalently, $[W_0, X]$ has full column rank for any basis W_0 of $\text{col}(W)$; the full set of unit and time indicators is itself rank-deficient by one and is not the object to check). This is a condition to be verified by computing the rank, since no simple combinatorial condition guarantees it: with event dates $E \in \{0, 2, 4\}$, periods $t \in \{4, \dots, 7\}$, and window $k \in \{2, \dots, 5\}$, every period contains several event times and every event time is observed in several periods, yet the eight columns of $[W_0, X]$ have rank seven, because all event dates are even and $k \bmod 2$ is then a function of t —the stride phenomenon of Theorem 1 again. Let ℓ be the affine function of k with $\ell(k_0) = \tau(k_0 + \delta)$ and $\ell(k_1) = \tau(k_1 + \delta)$. Then, conditional on the design, the coefficients on the reports and on the unconditioned outcomes satisfy, for every $k \in \mathcal{K} \setminus \{k_0, k_1\}$,

$$\mathbb{E}[\hat{\beta}_k - \hat{\beta}_k^\circ] = \tau(k + \delta) - \ell(k), \quad (6)$$

exactly. In particular, if the unconditioned outcome is such that $\mathbb{E}[\hat{\beta}_k^\circ] = 0$ for $k < 0$ —an additional premise, satisfied for instance under no anticipation, parallel trends, and dynamic effects that are homogeneous across event cohorts, but not implied by no anticipation and parallel trends alone when effects are heterogeneous across cohorts (Sun and Abraham 2021, secs. 3.4–3.5)—then the reported pre-period coefficients equal $\tau(k + \delta) - \ell(k)$: they are nonzero whenever τ is not affine on the pre-period tenure range, with no anticipation, no selection, and no violation of parallel trends in the outcome.

Proof. By Equation 3 the reported outcome is $Y = Y^\circ + v + \nu$ with $v_{it} = \tau(s_{it}) = \tau(k_{it} + \delta)$. Decompose $\tau(k + \delta) = \ell(k) + r(k)$ with $r(k_0) = r(k_1) = 0$. The affine part satisfies $\ell(k_{it}) = \ell_0 + \ell_1(t - E_i)$, the sum of a function of t and a function of i , hence $\ell(k) \in \text{col}(W)$ and $M\ell(k) = 0$. The remainder satisfies $r(k_{it}) = \sum_{k \in \mathcal{K} \setminus \{k_0, k_1\}} r(k) X_{k,it}$ because r vanishes at the omitted categories, so $Mv = MX\gamma$ with $\gamma = (r(k))_k$. By Frisch–Waugh–Lovell the event-time coefficients are $(X'MX)^{-1}X'MY$, and since MX has full column rank, $(X'MX)^{-1}X'Mv = \gamma$ exactly; D1 removes ν in expectation. $\square$

The theorem has two consequences that distinguish this artifact from the failures the pre-trends literature is built to catch. Outcome-side diagnostics may detect the resulting discrepancy without identifying its measurement origin: the unconditioned outcome satisfies parallel trends by construction, so a pretest (Roth 2022) may detect the discrepancy without identifying whether it arises from measurement drift, and it need not detect it at all, since power depends on the size of $\tau(k + \delta) - \ell(k)$ relative to sampling error. Sensitivity analyses that place restrictions on violations of parallel trends (Rambachan and Roth 2023) remain applicable if the measurement-induced deviation belongs to the restriction set chosen; what Equation 6 adds is the *shape* of the deviation, $\tau(k + \delta) - \ell(k)$, which is neither linear nor smooth in general and which changes with the choice of reference categories (the term ℓ moves with k_0, k_1), so that a restriction set chosen with outcome-side violations in mind may or may not contain it. Detecting a pre-trend is not identifying its cause; removing a measurement drift does not remove other confounding. The artifact is in principle correctable from measurement-side information, because Equation 6 is a deterministic transform of the conditioning path; but under exact alignment the correction cannot be made within the sample. Each event-time indicator is then a tenure indicator, $X_k = Z_{k+\delta}$, so the joint regression of Theorem 4 with tenure indicators added cannot separate event dynamics from conditioning: the residual-rank condition (R) fails by construction, and the sample’s own cell means identify only the curvature of the *sum* of the event path and τ . What is required is an *external* estimate of τ on the relevant tenure range—from a sample in which event time and tenure are not aligned, from a refreshment design of the kind treated in the companion paper, or from a shape library—whose identified part, $\tau(k + \delta) - \ell(k)$ for the analyst’s reference categories, is subtracted from the estimated coefficients. Simulation 4 reproduces Equation 6 to machine precision (Appendix C) and exhibits the rank-deficient design above; when MX is rank-deficient the decomposition of the drift between time effects and event-time coefficients is not unique and Equation 6 does not apply as stated.

Proposition 4 (Partial alignment). *If $s_{it} = k_{it} + \delta_i$ with δ_i varying across units, the drift is no longer a function of event time alone and the shift in the event-study coefficients is the*

linear projection

$$\mathbb{E}[\hat{\beta} - \hat{\beta}^\circ] = (X'MX)^{-1}X'M\tau(s) = (X'MX)^{-1}X'MZ_\tau\tau,$$

which reduces to Equation 6 when $\delta_i \equiv \delta$. For a given path the shift is zero if and only if $X'MZ_\tau\tau = 0$, and it is zero for every path if and only if $X'MZ_\tau = 0$: the residualized event-time indicators must be orthogonal to every tenure indicator. Balance of the tenure composition of each event-time category across calendar periods does not suffice: with eight event dates $E = 6, \dots, 13$, two units per date with $\delta_i \in \{6, 7\}$, window $k \in [-5, 6]$ and references $k_0 = -1$, $k_1 = -2$, every event-time category has half its observations at tenure $k + 6$ and half at $k + 7$ in every period, $[W_0, X]$ has full column rank 44, and yet the path $\tau(s) = 0.3\{1 - e^{-0.6(s-1)}\}$ shifts the pre-period coefficients at $k = -5, -4, -3$ by -0.142 , -0.054 , and -0.014 . When δ_i varies, the joint regression of Theorem 4 can be feasible, since X and Z_τ are no longer identical, and (R) is again the condition to check.

Proof. Frisch–Waugh–Lovell with $Z = [W, X]$ gives the projection formula, and $\tau(s) = Z_\tau\tau$. The zero conditions read off the formula: $(X'MX)^{-1}$ is nonsingular, so the shift vanishes iff $X'MZ_\tau\tau = 0$, and for every τ iff every column of $X'MZ_\tau$ vanishes. The example is verified in Appendix C; its point is that M residualizes on unit as well as time effects, so a tenure composition that is balanced within event time across *periods* is not thereby balanced after unit effects are removed. $\square$

Proposition 5 (Measurement-side sensitivity bounds for event studies). *Assume the conditions of Theorem 5 (exact alignment, $s = k + \delta$), with adjacent reference categories k_0 and $k_1 = k_0 - 1$, and suppose an external source—a staggered panel with stride one, a refreshment design, or substantive knowledge—bounds the curvature of the conditioning path on the tenures the window visits:*

$$|\Delta^2\tau(s)| \leq C \quad \text{for all } s \in [\min \mathcal{K} + \delta, \max \mathcal{K} + \delta - 2].$$

Then the measurement-induced shift of every event-study coefficient obeys

$$|\mathbb{E}[\hat{\beta}_k - \hat{\beta}_k^\circ]| = |\tau(k + \delta) - \ell(k)| \leq C \frac{m_k(m_k + 1)}{2}, \quad m_k = \begin{cases} k - k_0, & k > k_0, \\ k_1 - k, & k < k_1, \end{cases} \quad (7)$$

and the bound is attained when $\Delta^2\tau$ is constant at $\pm C$ with alternating sign relative to the weights. Two features make this operational. The shift is invariant to the affine direction of Θ_τ : it is unchanged when τ is replaced by $\tau + a + b(s - 1)$, because ℓ is the affine interpolant and absorbs the perturbation. Hence no normalization of the level of conditioning is required to apply Equation 7—only a bound on identified curvature. If the calibrating design has stride one, so that its kernel is the affine line, the shift is itself an identified functional of τ and Equation 7 can be replaced by a point correction; if the calibrating design has stride $d > 1$, its periodic directions are not affine in k , the shift is identified only up to them, and the bound must be widened by the corresponding periodic range.

Proof. Write $f(x) := \tau(x + \delta)$ and $L[f]$ for the affine function through $(k_0, f(k_0))$ and $(k_1, f(k_1))$, so that the shift is $f(k) - L[f](k)$ by Equation 6. The map $f \mapsto f(k) - L[f](k)$ annihilates affine functions, hence factors through second differences; with adjacent nodes, telescoping the first differences gives, for $k > k_0$,

$$f(k) - L[f](k) = \sum_{x=k_0}^{k-1} \{D(x) - D(k_1)\} = \sum_{x=k_0}^{k-1} \sum_{y=k_1}^{x-1} \Delta^2 f(y), \quad D(x) := f(x+1) - f(x),$$

so the weight on $\Delta^2 f(y)$ is the number of $x \in [k_0, k - 1]$ with $x > y$, namely $k - 1 - y$ for $y = k_1, \dots, k - 2$; these are $1, 2, \dots, m_k$ and sum to $m_k(m_k + 1)/2$. The case $k < k_1$ is symmetric. Invariance under $f \mapsto f + a + bx$ holds because $L[a + bx] = a + bx$. $\square$ $\square$

The bound is a ruler for reading a pre-trend. In the configuration of Simulation 4—references at $k_0 = -1$, $k_1 = -2$, window $k \in [-5, 6]$ —the constants $m_k(m_k + 1)/2$ are 6, 3, 1 at $k = -5, -4, -3$ and 1, 10, 28 at $k = 0, 3, 6$, so a path whose identified curvature is bounded by $C = .06$ in outcome units can move the pre-period coefficients by at most .37, .18, and .06, and the post-period coefficient at $k = 6$ by at most 1.7. The actual shifts there are .18, .07,

and .02: the bound is conservative by a factor of two to three, as it must be, since it allows the curvature to alternate in sign. Two practical readings follow. A researcher who finds pre-period coefficients larger than the bound knows that measurement drift of the maintained magnitude cannot account for them; one who finds them smaller cannot separate the two explanations without an external estimate of the path. And because the constants grow quadratically in the distance from the reference categories, the diagnostic value of distant leads and lags is exactly where the measurement artifact is largest—a reason to report the bound alongside the coefficients rather than to extend the window.

6.5 Which functionals survive

Proposition 6 (Orthogonality characterization). *A linear panel functional $\theta = \sum_{it} \lambda_{it} \mathbb{E}[Y_{it}]$ is unbiased for its unconditioned counterpart $\theta^\circ = \sum_{it} \lambda_{it} \mathbb{E}[Y_{it}^\circ]$ under every drift τ in a class $\mathcal{M}$ if and only if $\sum_{it} \lambda_{it} \tau(s_{it}) = 0$ for all $\tau \in \mathcal{M}$. For $\mathcal{M}$ the affine paths this is the pair of moment conditions $\sum_{it} \lambda_{it} = 0$ and $\sum_{it} \lambda_{it} s_{it} = 0$, satisfied by every estimator that partials out unit and time effects; for $\mathcal{M}$ unrestricted it is tenure-composition balance, $\sum_{it: s_{it}=s} \lambda_{it} = 0$ for every s in the support.*

Proof. $\theta - \theta^\circ = \sum_{it} \lambda_{it} \tau(s_{it})$ by Equation 3 and D1; the “if and only if” is the definition of unbiasedness for all $\tau \in \mathcal{M}$. For affine $\mathcal{M}$ the condition is linear in (a, b) and reduces to its two coefficients; for unrestricted $\mathcal{M}$ take $\tau = \mathbf{1}\{s = s_0\}$ for each s_0 . The TWFE weights satisfy the affine pair by the proof of Theorem 3. $\square$

The second condition is the bridge to the design problem treated in companion work: a published aggregate that is balanced on time in sample is a linear functional satisfying the composition condition, so it is drift-free, and *the same functional* carries no information about the drift. This is a statement about one functional at a time—a robust aggregate estimator does not imply that the survey’s rotation-group-level data are uninformative about conditioning, only that the aggregate is; the companion paper’s balance–information theorem makes the distinction precise.

7 Monte Carlo Design (ADEMP)

Aims: noiseless verification of the identities of Theorem 1 and Corollary 1 on the JLPS-calibrated support (stride $d = 4$); size and power of the plateau observable-implications test; coverage of the drift-bound intervals of Proposition 3 at the covered tenures; dependence of level estimates on the normalization. *Data-generating mechanisms:* 19-wave staggered panel calibrated to JLPS (three cohorts, entry at $t = 1, 5, 13$); conditioning paths (null, saturating, linear-drift, non-monotone); cohort effects (null, drifting, shock at 2011 entry). *Estimands:* the lag-4 second difference $\Delta_4^2\tau(1)$ and the identified contrast $\Delta^2\tau(2) - \Delta^2\tau(6)$; $\bar{\tau} - \tau(1)$ under the plateau restriction; drift bounds for $\tau(13) - \tau(1)$ and $\tau(17) - \tau(1)$ at $M \in \{0.4, 1, 2, 4\} \times$ true maximal drift. *Methods:* explicit cell-mean contrasts and regression readouts from arbitrary representatives; the plateau-constrained estimator with its observable-implications test; drift-bound intervals with Imbens–Manski expansion. *Performance measures:* bias, MCSE, identified-set coverage, specification-test rejection. Simulation 4 (Appendix C) verifies the identities of Section 6 (absorption, invariance, the identified correction, and the exact pre-trend formula) to machine precision and reports the finite-sample behaviour of the corrected estimator. The full protocol, with prespecified versus exploratory analyses flagged, is Appendix B (ADEMP-PreReg format); results are in Appendix A.

8 Illustration: A 19-Wave Panel with Two Refreshment Samples

This section illustrates the design calculus; it reports no new empirical estimates. The Japanese Life Course Panel Survey (JLPS) enrolled cohorts in 2007, 2011, and 2019—waves 1, 5, and 13—and has been observed through wave 19. Its stride is $d = 4$, its second cohort entered $\Delta_2 = 4 = d$ waves after the first, so Lemma 1(i) holds for every T and C_4 is satisfied; its cell design has 41 cells, 39 free parameters, rank 35, and a four-dimensional kernel, exactly the affine-plus-periodic family. By Corollary 1 the curvature of conditioning is identified only at lag four, and the drift bounds of Proposition 3 cover tenures 5, 9, 13, and 17. These are precisely the tenures at which its refreshment samples enter and at which its increments

are estimated, so the design is well matched to the objects the companion papers estimate and poorly matched to the ordinary second differences an analyst might have expected to read off it. Two features of the design bear on Section 5. The entry negative-control set is unusually rich—23 time-invariant childhood-circumstance items asked once, at entry, of every cohort—so the affine direction can be anchored under the conditions of Proposition 2; but with $d = 4$ the battery leaves the three periodic directions untouched, and the path at tenures $s \not\equiv 1 \pmod{4}$ is recovered only under the plateau restriction of Proposition 1 or an external anchor. And the 2019 entrants share no birth-cohort support with the incumbents, so that episode anchors increments rather than levels—the configuration treated fully in the companion refreshment-theory paper. The companion empirical paper implements these recovery conditions on the restricted microdata under the archive’s disclosure rules and reports the resulting battery contrasts and conditioning estimates; we do not reproduce those numbers here, and the present paper’s claims do not depend on them.

8.1 An illustration from published rotation-group indices

Theorem 2 says that a rotation design’s month-in-sample differences identify the per-month rates of conditioning up to a common constant. Nothing about that statement requires microdata: the rates are computable from the month-in-sample indices that statistical agencies have published for half a century. This subsection computes them, as an illustration of what the identified object looks like in practice and of how the assumptions announce their own failure.

Write $b(k)$ for the published index at month in sample k and $y(k) = \log b(k)$, since the published indices are ratios to a benchmark and the additive model of Equation 2 applies to the logarithm; over the ranges involved the distinction is numerically immaterial. With the CPS pattern $j = (0, 1, 2, 3, 12, 13, 14, 15)$, the identified object is the profile of per-month rates

$$r_k = \frac{y(k+1) - y(k)}{j(k+1) - j(k)}, \quad k = 1, \dots, 7,$$

determined up to adding the same constant to all seven (Theorem 2(c)); equivalently, the six differences $r_k - r_l$ are point identified, and the level of the profile is not. Two hypotheses are

testable against it. *Conditioning accumulates with calendar time*: all seven rates are equal. *Conditioning accumulates with interviews at a constant rate*: the six within-wave rates are equal and the rate across the rest period is one ninth of them.

Table 2: Per-month rates of conditioning implied by published CPS month-in-sample indices, in log points per month, with $\mathcal{D} = [\tau(5) - \tau(4)] - 9[\tau(2) - \tau(1)]$ of Corollary 2. The rates are identified up to a common additive constant; $\mathcal{D}$ is identified outright and is zero for any path that accumulates with calendar time. [†] In 1968–69 the CPS asked additional questions of persons not in the labor force *only* at months in sample 1 and 5 (Bailar 1975), so this row’s instrument varies with the dose and assumption M1 fails; it is shown to make the failure visible.

Source and item	r_4							$\mathcal{D}$
	r_1	r_2	r_3	(rest)	r_5	r_6	r_7	
McIllece (2022, Tab. 2): unemployed, 2003–22	−.055	−.038	−.025	+ .001	−.046	−.017	−.001	+ .510
McIllece (2022, Tab. 2): employed, 2003–22	.000	−.001	−.005	+ .000	−.010	+ .001	+ .003	+ .002
Bailar (1975, Tab. 1): hours 35–40, 1968–69	+ .052	+ .022	+ .016	−.003	+ .032	+ .011	+ .008	−.502
Bailar (1975, Tab. 1): hours 35–40, 1970–72	+ .048	+ .022	+ .016	−.003	+ .025	+ .014	+ .008	−.457
Bailar (1975, Tab. 1): unemployed, 1968–69 [†]	−.167	−.052	−.038	+ .018	−.125	−.041	−.017	+ 1.671

Three readings. First, the calendar clock is decisively rejected wherever the index moves at all: for unemployment over 2003–22 the rates range over .057 log points per month and $\mathcal{D} = +.51$, against zero under that hypothesis. Second, the constant-per-interview clock fares no better, and the reason is visible in the profile: within each four-month wave the rate attenuates (−.055, −.038, −.025 in the first wave; −.046, −.017, −.001 in the second), and the second wave’s first rate is nearly as steep as the first wave’s. A path that depends on the cumulative number of interviews cannot produce a steeper marginal effect at the fifth interview than at the third. The rest period appears to restore part of the responsiveness that

the first wave used up—which is the “pattern ... also apparent within each four-month wave” that McIllece (2022) reports, given identification content here: the within-wave attenuation and its resumption are differences of rates, hence normalization-free, whereas the levels that a reader is tempted to compare across waves are not. Third, the employment index illustrates the negative case. Its rates are an order of magnitude smaller and $\mathcal{D} = +.002$; the design has almost nothing to detect, and a paper reporting an employment month-in-sample “bias” of a tenth of a percent is reporting a quantity whose sign is normalization-dependent.

The last row of the table is a caution, not a finding. Because the 1968–69 CPS asked additional labor-force probes only of households in their first and fifth months in sample, the instrument varied with the dose—a direct violation of the clause of M1 that all cohorts face the same instrument at a given time—and the resulting profile shows it: the two intervals that step off or onto a probed month in sample, $1 \rightarrow 2$ and $4 \rightarrow 5$, are precisely the outliers—the steepest rate in the table and the only positive one. Solon (1986, Tab. 1) shows the same artifact after the probes were moved to months in sample 4 and 8 in 1970: in his 1974–83 averages the only two positive within-wave rates are the intervals $3 \rightarrow 4$ and $7 \rightarrow 8$, which are exactly the two that end at a probed month. The framework does not repair such a design, but it does localize the damage, and it says which contrasts survive it: any functional whose weights avoid the affected months and annihilate $j(\cdot)$, for instance the comparison of the rate from month three to month five with the rate from month two to month three, which in Solon (1986)’s averages differ by 14.9 thousand persons per month.

9 Discussion

What the theorems change in practice. Four practical implications follow. First, any reported *level* of panel conditioning or rotation-group bias should be read jointly with its normalization; comparisons of levels across studies that normalize differently are comparisons of conventions, not of surveys. Our genealogy table makes the conventions explicit for the major published estimates. Second, the *identified shape* of conditioning—curvature at the design’s stride, and ordinary curvature when entry cohorts are adjacent—is the exportable, normalization-free quantity; meta-analyses and bias-prior libraries should accumulate iden-

tified contrasts rather than level estimates without their identifying restrictions, and should record the stride of the design that produced each one. Third—and this is new to the downstream results—fixed-effects users can estimate Equation 5, the two-way regression with tenure indicators, rather than either ignoring conditioning or choosing a level; event-study users on tenure-aligned designs should treat pre-period coefficients as suspect until the measurement-side check of Theorem 5 has been run. Fourth, rotation designs should report month-in-sample differences as a *dose response against elapsed time*: the rest period between a rotation group’s interview blocks is what makes the accumulation clock visible (Theorem 2), and the contrast of Corollary 2 is computable from figures official statistical agencies already publish. Fifth, before fielding or analyzing a staggered design, compute the kernel of its cell design matrix (Theorem 1(a), (d)): the refreshment schedule and the follow-up of its last cohort fix which functionals of conditioning the panel can ever deliver, a schedule with a common divisor larger than one forfeits ordinary curvature, and a cohort followed too briefly forfeits more (Lemma 1).

Relation to companion work. The identification theory of refreshment designs under attrition—the sharp identified set under a single refreshment, and conditions for point identification with refreshment samples at multiple tenures—is developed in companion work, where the JLPS’s two refreshment cohorts (2011, 2019) with a repeated entry-wave negative-control battery constitute, to our knowledge, the first dataset satisfying those conditions. The downstream question for *additive* tenure drift is settled in Section 6; its extension to report-dependent (state-dependent) measurement error, where the drift depends on the previous report and is not absorbed by fixed effects, is the subject of a companion paper on transition estimators. The same companion work develops the design problem: which rotation, refreshment, and mode-transition schedules make conditioning identifiable at all, and at what cost—including the result that rotation designs balanced on time in sample hold the tenure component of published aggregates constant over time at the price of the information from which it could have been estimated.

Limits. The model is additive throughout, in two places: cell means are additive in period, cohort, and conditioning (M4), and the individual report is the unconditioned outcome plus a

common path (D1). Both are the standard assumptions of the rotation-group-bias and panel-conditioning literatures, and both are where these results end. With cohort-by-tenure or period-by-tenure interactions the indeterminacy is item-wise or cohort-wise and larger; with conditioning that depends on the previous *report* rather than on the count of interviews—state-dependent measurement error—the drift is not a function of the design alone, two-way fixed effects do not absorb it, and the companion paper on transition estimators takes it up. Nothing here is a general theory of measurement error, and the results should be read as statements about an additively separable conditioning path; Solon (1986) argued, from the rotation-group evidence itself, that the additive model of rotation-group bias is not consistent with the data for unemployment, and proposed a multiplicative alternative—on published multiplicative indices the natural reading of our model is that it applies to the logarithm of the index, which is what we do in Section 8.1. Our impossibility results are about nonparametric identification from response distributions alone; they do not bind designs with auxiliary information (validation records, paradata with exclusion restrictions, randomized question exposure), which is precisely where the recovery conditions point. And the additive cell-mean structure, while standard, is a choice: with item-by-tenure interactions the indeterminacy is item-wise, which strengthens rather than weakens the case for shape-based reporting.

Related literature

Our results connect four literatures that have proceeded in parallel: the rotation-group-bias tradition in official statistics (Bailar 1975; Solon 1986; Krueger, Mas, and Niu 2017; McIllece 2022), where normalizations are ubiquitous and the identification obstacle is folklore—and where, by Theorem 2, the interrupted rotation pattern that tradition works with is precisely what makes its month-in-sample differences readable as a response to interviews rather than to time; the panel-conditioning literature in survey methodology (Warren and Halpern-Manners 2012; Halpern-Manners and Warren 2012; Halpern-Manners, Warren, and Torche 2017; Kraemer et al. 2024, 2025), which documents effects, and within which Das, Toepoel, and van Soest (2011) give a formal identification analysis of conditioning and attrition for a

binary item in a two-wave design with a refreshment sample—the closest antecedent to our Section 4 and the companion refreshment paper; the APC identification literature (Mason et al. 1973; Fosse and Winship 2019a, 2019b), whose linear dependency we transplant to the measurement domain, and whose unequal-interval extension (Gascoigne and Smith 2023) already contains the periodic ambiguity that a common refreshment stride produces; and the modern event-study literature (Sun and Abraham 2021; Borusyak, Jaravel, and Spiess 2024), whose fully dynamic underidentification and contamination results have exact analogues here—with tenure in the role of event time and the survey instrument in the role of the outcome equation. Feng, Hu, and Sun (2022) point-identify CPS misclassification probabilities in a latent-state model with history-dependent errors; their object and model class differ from the additive mean model here, and we make no claim that their restrictions select a point of Θ_τ . What the two share is the dependency between tenure, period, and cohort that both must confront; the entry-wave negative controls used here check the comparability of entering groups on time-invariant facts, one component of the cross-rotation-group comparability their transport step requires, and not that condition in full. Bertoli, Jakli, and Pascoe (2026) formalize conditioning bias in two-wave event-impact surveys and propose a dual-randomization design; our results cover the many-wave staggered case that their design decisions instantiate.

Appendix A: Simulation results

All simulations use the JLPS-calibrated support—cohorts entering $t = 1, 5, 13$, $T = 19$, stride $d = 4$ —so that the identified set of Theorem 1 is four-dimensional and ordinary second differences are not identified. The specifications below are those of the code, which is definitive; every number quoted in this appendix is read from a named column of the corresponding output file, and the replication package’s `check_manuscript_values.R` reproduces each of them from those files.

Simulation 1: noiseless identities, normalizations, and the plateau test

Design: $B = 500$; cell noise $1/\sqrt{1000}$ (one thousand respondents per cell with unit outcome variance); scenarios S1 null / S2 saturating, $\tau(s) = 0.2\{1 - e^{-0.6(s-1)}\}$ / S3 = S2 + linear drift $0.01(s - 1)$ / S4 = S2 + late nonlinear drift $0.004\max(s - 6, 0)^2$; cohort shock G1 (+0.15) on the 2011 cohort; plateau window $s \geq 6$.

Noiseless identity tests. On population cell means, adding the periodic null direction $\rho = (0, 0.30, -0.10, 0.20)$ (period four, $\rho(1) = 0$) to τ and subtracting $\rho(t)$ from α changes no cell mean (5.6×10^{-17}). The ordinary second difference $\Delta^2\tau(3)$ moves from -0.022 to $+0.678$; the lag-4 second difference $\Delta_4^2\tau(1) = \tau(9) - 2\tau(5) + \tau(1)$ does not move; and the explicit cell-mean contrast $[\mu(1, 9) - \mu(5, 9)] - [\mu(1, 5) - \mu(5, 5)]$ —first argument the entry date, as in Equation 1; the cohorts entering at $t = 1$ and $t = 5$ compared at $t = 9$ and at $t = 5$ —returns $\Delta_4^2\tau(1) = \tau(9) - 2\tau(5) + \tau(1) = -0.1654$ to 10^{-17} before and after the perturbation. (The same two cohorts compared at $t = 13$ and $t = 9$ would return $\Delta_4^2\tau(5) = -0.0150$, a different identified functional.) This is Theorem 1 and Corollary 1 at machine precision.

	bias level	bias level	$\Delta_4^2\tau(1)$	bias (con-		bias $\bar{\tau}$	
Scenario	N1	N2	true	trast)	MC sd	(plateau)	rejection
S1 null	0.000	0.000	0.000	0.001	.064	0.001	.058
S2 satu-	-0.051	-0.102	-0.165	0.004	.061	-0.006	.048
rating							
S3 +	-0.089	-0.140	-0.165	0.001	.065	-0.097	.054
linear							
drift							
S4 +	-0.173	-0.124	-0.129	0.005	.060	-0.408	1.000
nonlinear							

Rows are the G0 (no cohort shock) setting; the G1 rows of `sim1_results.csv` differ from them only within Monte Carlo error. Three theorem-consistent facts: (i) level estimates

differ by normalization; (ii) the identified lag-4 curvature, estimated by an explicit contrast, is unbiased in every scenario (Monte Carlo standard error .003); (iii) the plateau *observable-implications* test (constancy over time of the cohort gaps within the window) has correct size under the null and under a pure linear drift—the violation Proposition 1 declares undetectable—and power one against the nonlinear violation, while the plateau-constrained estimator absorbs the drift it cannot see (−0.097 under S3). [sims/sim1_normalizations.R; sim1_results.csv]

Simulation 2: identified and unidentified curvature readouts

The target is $\theta := \Delta^2\tau(2) - \Delta^2\tau(6)$, an identified functional on this support—its weights sum to zero within every residue class modulo four and are orthogonal to $s - 1$ —contrasted with $\Delta^2\tau(3)$, which is not identified (Corollary 1). A saturated regression returns an arbitrary representative of Θ_τ , from which unidentified functionals read arbitrarily and identified ones read exactly.

Noiseless identities. $\theta = -0.03702$; the correctly centred contrast $-\Delta^2D(6)$, $D(t) = \mu(1, t) - \mu(2, t)$, returns it to 10^{-16} ; the regression readout $\Delta^2\hat{\tau}(2) - \Delta^2\hat{\tau}(6)$ from an arbitrary representative returns it to 10^{-15} ; the regression readout of $\Delta^2\hat{\tau}(3)$ is -0.072 against a true -0.022 , and moves with the representative.

Scenario	θ true	bias (con-		bias (re-		$\Delta^2\tau(3)$ true	mean readout
		trast)	MC sd	gression readout)	MC sd		
S2	−0.037	0.006	.110	0.000	.080	−0.022	−0.075
S3	−0.037	0.003	.107	0.003	.081	−0.022	−0.072

Both estimators of the identified functional are unbiased (Monte Carlo standard error .005); the regression readout is the more efficient because it uses every cell. The recommendation: read identified functionals off any representative, but check identification first. [sims/sim2_direct_contrast.R; sim2_results.csv]

Simulation 3: drift-bound intervals (Proposition 3)

Design: same support; saturating $\tau(s) = 0.25\{1 - e^{-(s-1)/2}\}$; cohort effects $g = (0, 0.10, 0.22)$, per-period drifts $(0.025, 0.015)$, so the true maximal drift is $M_0 = 0.025$; $n = 500$ per cell, $B = 300$, endpoint standard errors by parametric bootstrap (150 draws). The interval is $[\hat{\tau}_{lo} - c\hat{\sigma}_{lo}, \hat{\tau}_{hi} + c\hat{\sigma}_{hi}]$ with the critical value c of Imbens and Manski (2004) solving $\Phi(c + \hat{\Delta}/\hat{\sigma}) - \Phi(-c) = 0.95$, $\hat{\Delta}$ the estimated width of the identified set and $\hat{\sigma}$ the larger endpoint standard error, so that $c \rightarrow 1.645$ for a wide set and $c \rightarrow 1.96$ for a point-identified one; the target is coverage of the true parameter value, not of the whole set. The bounds are evaluated at $s = 13$ and $s = 17$, the tenures congruent to 1 modulo 4 at which Proposition 3 applies. A second design, G3, sets the two true drifts to $(+0.025, -0.025)$, so that at $M = M_0$ both constraints bind and the identified set for m is a single point; it is included to check the interval where the set is narrow.

At the population level the representative's implied drifts under G2 are $(0.0117, 0.0017)$, each displaced from the truth by the same constant—the linear part the representative reassigns to the cohort dimension—while their difference is preserved; on the subgrid $s \in \{1, 5, 9, 13, 17\}$ the representative agrees with the truth up to that linear part to 10^{-16} . At $M = M_0$ the sharp interval for $\tau(13)$ is $[-0.23, 0.25]$ and contains the truth 0.249 at its upper endpoint (the true drift equals M_0); at $M = 0.4M_0$ it is $[-0.05, 0.07]$ and excludes it.

DGP	M/M_0	coverage, $\tau(13)$				coverage, $\tau(17)$		
		(IM / fixed z)	median width	set width	median c	(IM / fixed z)	median width	empty rate
G2	0.4 (vio- lated)	.127 / .123	0.33	0.12	1.65	.107 / .107	0.43	.05–.07
G2	1.0	.957 / .957	0.69	0.48	1.64	.957 / .957	0.91	0
G2	2.0	1.000 / 1.000	1.29	1.08	1.64	1.000 / 1.000	1.71	0

DGP	M/M_0	coverage, $\tau(13)$				coverage, $\tau(17)$		
		(IM /	median	set	median	(IM /	median	empty
		fixed z)	width	width	c	fixed z)	width	rate
G2	4.0	1.000 / 1.000	2.49	2.28	1.64	1.000 / 1.000	3.31	0
G3 (tied)	0.4 (vio- lated)	.020 / .007	—	−0.36	1.96	.023 / .007	—	1.00
G3 (tied)	1.0	.953 / .900	0.25	0.00	1.93	.940 / .910	0.33	.47–.50
G3 (tied)	2.0	1.000 / 1.000	0.82	0.60	1.64	1.000 / 1.000	1.08	0
G3 (tied)	4.0	1.000 / 1.000	2.02	1.80	1.64	1.000 / 1.000	2.68	0

Coverage at the true M is at the nominal level, conservative above it, and collapses under a violated bound. On G2 the identified set is wide relative to the endpoint standard errors, the Imbens–Manski critical value is 1.645, and the two intervals coincide. On G3 the identified set for m is a point at $M = M_0$, the estimated set is empty in half the replications (its estimated width is negative whenever sampling error puts the two binding constraints in the wrong order), the critical value rises to 1.93–1.96, and coverage of the true value is .953 and .940 (Monte Carlo standard error .013) against .900 and .910 with the fixed 1.645: the pointwise expansion loses five percentage points exactly where the set is narrow, as the theory of Imbens and Manski (2004) says it must. The intervals are computed for the true parameter value, not for the whole identified set; the endpoint standard errors are bootstrap standard deviations of the estimated endpoints, which are minima and maxima of finitely many normal statistics and are asymptotically normal only when one constraint binds strictly—a caveat that the tied design G3 makes visible and that a uniform treatment would address by resampling the constraint set. Two lessons attach. The intervals are wide at late tenure: the interval for m enters $\tau(s)$ multiplied by $(s - 1)$, so at $s = 17$ even the

correctly specified bound spans nearly a standard deviation; the level of long-run conditioning is expensive, and only some curvature contrasts are estimable, under the support conditions of Corollary 1. And with three cohorts the emptiness test has little power against modest violations: refutation of a drift bound needs more refreshment cohorts or an external anchor.

[sims/sim3_drift_bounds.R; sim3_results.csv]

Appendix B: Simulation protocol (ADEMP-PreReg format)

This appendix states the simulation protocol in the ADEMP structure, with the prespecification status of every analysis flagged: **[P]** = prespecified before any simulation was run; **[E→C]** = discovered in an exploratory run, then confirmed in a fresh simulation; **[R]** = added or corrected in response to the first review round; **[R2]** = in response to the second.

Aims. (A1) **[P]** Verify numerically that level estimates of τ depend on the normalization while identified curvature does not (Theorem 1, Corollary 1). (A1') **[R]** Noiseless identity tests of the periodic null direction. (A2) **[P]** Size and power of the plateau observable-implications test (Proposition 1), including its predicted blindness to pure linear drift. (A3) **[P]** Bias of the plateau-constrained estimator under correct and violated saturation. (A4) **[P]** Coverage of the drift-bound intervals (Proposition 3) at correctly specified, conservative, and violated M , at the covered tenures. (A5) **[R]** Compare readouts of identified and unidentified functionals from saturated regressions with explicit contrasts. (A6) **[R2]** Coverage of the Imbens–Manski interval against a fixed- z interval on a design in which the identified set is a point.

Data-generating mechanisms. JLPS-calibrated support: cohorts entering $t = 1, 5, 13$, $T = 19$. Sims 1–2: cell noise $1/\sqrt{1000}$; paths S1 null, S2 $\tau(s) = 0.2\{1 - e^{-0.6(s-1)}\}$, S3 = S2 + 0.01($s - 1$), S4 = S2 + 0.004 $\max(s - 6, 0)^2$; cohort effects G0 null, G1 shock +0.15 on the 2011 cohort. Sim 3: outcome sd 1, $n = 500$ per cell; S2': $\tau(s) = 0.25\{1 - e^{-(s-1)/2}\}$; G2 drifting $g = (0, 0.10, 0.22)$, $M_0 = 0.025$; G3 tied drifts $g = (0, 0.10, -0.10)$, drifts $(+M_0, -M_0)$. Factorial: sim 1 crosses S1–S4 $\times$ {G0, G1}; sim 2 runs S2, S3 $\times$ G1; sim 3

runs $S2' \times \{G2, G3\}$. $B = 500$ (sims 1–2), $B = 300$ with 150 bootstrap draws (sim 3).

Estimands. $\Delta_4^2\tau(1)$ and $\theta = \Delta^2\tau(2) - \Delta^2\tau(6)$ (identified); $\Delta^2\tau(3)$ (not identified; reported to exhibit its arbitrariness); $\bar{\tau} - \tau(1)$ under the plateau restriction with window $s \geq 6$; $\tau(13) - \tau(1)$ and $\tau(17) - \tau(1)$ for the drift bounds; the observable-implications and emptiness test statistics.

Methods. Saturated cell-mean OLS under normalization N1 (reference level) and N2 (zero mean along the affine direction); explicit cell-mean contrasts; the plateau-constrained estimator with the cohort-gap-constancy test; the representative-plus-drift-interval construction of Proposition 3 with the linear part read on the subgrid $s \equiv 1 \pmod{4}$, parametric-bootstrap endpoint SEs, and the interval $[\hat{\tau}_{lo} - c\hat{\sigma}_{lo}, \hat{\tau}_{hi} + c\hat{\sigma}_{hi}]$ with the critical value c of Imbens and Manski (2004) (coverage of the true parameter value; c between 1.645 and 1.96 according to the estimated width relative to the endpoint standard error), with the fixed- z (1.645) interval reported alongside.

Performance measures. Bias and Monte Carlo SE of each estimand; exact invariance of identified functionals under null-direction perturbations (predicted identity, not approximation); rejection rates of the observable-implications test at nominal .05; coverage of the true $\tau(s)$ by the Imbens–Manski interval and by the fixed- z interval, median interval width, median estimated set width, and median critical value at each $M/M_0 \in \{0.4, 1, 2, 4\}$; emptiness rate.

Deviations from protocol. Recorded here in full; the response letters to the two review rounds give the history. (1) Proposition 3 was restated in terms of first-difference drift, which the affine direction moves, before sim 3 was run; an earlier draft had bounded second differences, which it does not. (2) Sims 1–2 were rewritten with noiseless identity tests after the periodic null direction was recognized; an earlier reading of the same discrepancy as “boundary-dummy contamination” of saturated regressions was a misreading of the non-identification of ordinary second differences on a stride-four support and is withdrawn. (3) Sim 2’s contrast estimator is centred so that its estimand is θ above. (4) [R2] Sim 3 replaced a fixed critical value of 1.645 by the Imbens–Manski critical value and added the tied-drift design G3, on which the fixed value gives 90% coverage. (5) [R2] Sim 4’s attrition mechanism

was made monotone, consistent with D2; the design of the exactly aligned event study is unchanged. (6) [R2] Sim 1’s cell-mean contrast for $\Delta_4^2\tau(1)$ compares the two cohorts at $t = 9$ and $t = 5$, as the code always did; the appendix prose had named the wrong periods. All scripts and seed-controlled outputs accompany the replication package, and `check_manuscript_values.R` reproduces every number quoted in Appendices A and C from named output columns.

Appendix C: Simulation 4 — downstream propagation

Design. Staggered panel with cohorts entering $t = 1, 5, 13$ (stride $d = 4$), $T = 19$, $n = 200$ per cohort, unit and time effects, monotone attrition with an exit hazard increasing in tenure (a unit that leaves does not return, so that D2 holds and the support is an unbalanced trapezoid), and a staggered binary regressor D adopted at a random calendar time after entry (30% never adopt) with true coefficient 0.5. Reports add a conditioning path to the unconditioned outcome: S (saturating, $\tau(s) = 0.30\{1 - e^{-0.6(s-1)}\}$), L (affine, $0.05(s - 1)$), or S+L. For the pre-trend block, eight consecutive cohorts ($e = 1, \dots, 8$, $T = 20$, $n = 60$, no attrition) each experience an event five waves after entry ($\delta = 6$), the estimation window is $k \in [-5, 6]$, and the unconditioned outcome has no event effect and no anticipation; the fully dynamic specification omits $k_0 = -1$ and $k_1 = -2$. Seed 20260916; code `sims/sim4_downstream.R` (v3); every number below is written by the script to `sim4_manuscript_values.csv` under a named quantity. The cell-mean design of the main support has 41 cells, 39 free parameters, rank 35, and nullity 4, as Theorem 1(d) requires for $d = 4$ under C_4 .

Deterministic identities. Conditional on one realization of the design, Theorem 3 holds exactly: the observed bias of the TWFE coefficient equals $(D'MD)^{-1}D'M\tau(s)$ to 2×10^{-14} (S and S+L: bias 0.0256; L: bias 9×10^{-15}). Adding the affine direction $0.7 + 0.2(s - 1)$ changes the estimate by 4×10^{-14} , and adding the *periodic* direction $\rho = (0, 0.30, -0.10, 0.20)$ of period four changes it by 9×10^{-15} : the bias is invariant along the entire four-dimensional identified set, as Theorem 3(i) states. In a single-cohort panel the bias is 10^{-16} for the saturating path (Corollary 3(a)). Under exact alignment, all ten event-study coefficients shift by $\tau(k + \delta) - \ell(k)$ to 1.5×10^{-14} (Theorem 5). The measurement-induced *shifts* in the

three pre-period coefficients are -0.183 , -0.070 , and -0.018 at $k = -5, -4, -3$; these are not the estimated coefficients themselves, which in this realization are $+0.068$, $+0.061$, and $+0.014$ on the reports and $+0.252$, $+0.131$, and $+0.032$ on the unconditioned outcome—the latter being sampling error of a regression with 480 units and no event effect, and the former that error plus the shift. On the same aligned design the residualized event-time and tenure indicators are collinear, $\text{rank}[MX, MZ_\tau] = 10$ against 21 columns, so condition (R) fails and no within-sample correction exists. The rank-deficient design of Section 6.4 (event dates $\{0, 2, 4\}$, periods 4–7, window 2–5, two references omitted) has eight columns and rank seven. The partial-alignment design of Proposition 4 (event dates $6, \dots, 13$, two units per date with $\delta_i \in \{6, 7\}$) has a tenure composition of exactly one half at each of $k + 6$ and $k + 7$ in every (k, t) cell, $\text{rank}[W_0, X] = 44$ of 44, $\max |X' MZ_\tau| = 6.0$, and shifts of -0.1419 , -0.0544 , -0.0142 at $k = -5, -4, -3$ for the saturating path.

Identified correction. On the main design, the joint regression Equation 5 (four tenure indicators aliased with the fixed effects and dropped) returns 0.578 against the infeasible unconditioned estimate 0.568 and the reported-outcome estimate 0.594, with unit-clustered standard error 0.051. Under tenure-determined adoption, $D = \mathbf{1}\{s \geq 4\}$, the regressor is collinear with the tenure indicators, condition (R) fails, and the joint regression reports no coefficient on D .

Monte Carlo. $B = 100$ replications, $n = 120$ per cohort, unit-clustered 95% intervals. Five designs: baseline (S+L path, random adoption); zero conditioning ($\tau \equiv 0$); heterogeneous treatment effects ($\beta_i \sim N(0.5, 0.2^2)$); cohort-dependent adoption (never-adopter shares .5, .3, .1 by cohort); tenure-determined adoption.

Design	bias,			bias,		
	reported	RMSE	coverage	corrected	RMSE	coverage
baseline	0.025	.067	.95	0.002	.063	.98
zero condi-	0.010	.063	.99	0.012	.065	.98
tioning						
heterogeneous	0.018	.059	.99	-0.006	.058	.99
effect						

Design	bias,	RMSE	coverage	bias,	RMSE	coverage
	reported			corrected		
cohort- dependent adoption	0.042	.074	.92	−0.001	.064	.98
tenure- determined adoption	0.149	.164	.46	not identified (R fails)	—	—

Monte Carlo standard errors of the reported biases are .006. The corrected estimator is unbiased in every design in which (R) holds and is unaffected by treatment-effect heterogeneity that is independent of the design, as Theorem 4 requires; its RMSE is below the naive estimator’s when conditioning is present (.063 against .067; .064 against .074) and marginally above it when there is none (.065 against .063), which is the observed price of estimating a tenure component that is zero. Its intervals are conservative here (.98–.99). Under tenure-determined adoption the naive estimator is badly biased and the corrected one does not exist—the design cannot separate exposure from conditioning, and the joint regression says so rather than returning a number.

Appendix D: Deterministic design checks

The results of Section 3 and Proposition 5 are algebraic, and the replication package checks them by exact computation rather than by simulation (`sims/check_interrupted.py`, `sims/check_lemma1_trapezoid.py`, `sims/design_rank.R`; rational arithmetic throughout, no randomness and no data).

Dose designs (Theorem 2, Corollary 2, Table 1). For each schedule, the script builds the cell design of Equation 2, computes its exact rank and a basis of the kernel, and reports whether the kernel’s τ -component is spanned by $j(\cdot)$ and whether a dose-linear path lies in it. With the CPS pattern $J = \{0, 1, 2, 3, 12, 13, 14, 15\}$ and monthly entry the design has nullity

one and kernel $\propto (0, 1, 2, 3, 12, 13, 14, 15)$ —that is, $j(k)$ —and the dose-linear path is *not* in the kernel; with $J = \{0, \dots, 7\}$ the kernel is $\propto (0, \dots, 7) = k - 1$ and the dose-linear path is in it, as Theorem 2(d) requires. The pattern $\{0, 1, 4, 5\}$ behaves like the CPS one and the equally spaced pattern $\{0, 2, 4, 6, 8, 10\}$ like the uninterrupted one. With entry every second period the CPS pattern gives nullity two, the second kernel direction being $\rho(j) = j \bmod 2$, the periodic direction of Theorem 2(a). Condition (P) binds: with the CPS pattern the nullity is two for entry windows of nine cohorts or fewer and one from ten cohorts on, ten being one more than the pattern’s largest gap. The equally spaced pattern $\{0, 2, 4, \dots\}$ also illustrates the incidence-graph caveat of Theorem 1(d): cohorts of odd and even entry never share a period, the incidence graph has two components, and the design carries a kernel direction that moves α and g against each other with no movement in τ at all.

Lemma 1 (o) and (iv). A sweep of all entry sets $\mathcal{E} \subset \{1, \dots, 13\}$ with $e_1 = 1$, $3 \leq K \leq 5$, and $T \leq e_K + 13$ —10,934 designs—finds no violation of $\nu = \#\{\text{increment-graph components}\}$ and none of $\nu \leq \max\{d, \Delta_2 - w\}$; the bound is attained in 9,746 of them and strict in 1,188 (`check_lemma1_components.py`, which repeats the sweep standalone). The version of the bound computed from the *third* cohort’s follow-up instead of the last cohort’s fails in 72 of these designs, the smallest being $\mathcal{E} = \{1, 5, 9, 11, 12\}$ with $T = 12$.

Sensitivity weights (Proposition 5). For the configuration of Simulation 4 the script solves for the representation of the shift functional in the basis of second differences and reports the weights, which are the ramps $\{1\}$, $\{1, 2\}$, $\{1, 2, 3\}$ at $k = -3, -4, -5$ and $\{1, \dots, 7\}$ at $k = 6$, with absolute sums 1, 3, 6 and 28—the triangular numbers $m_k(m_k + 1)/2$ of Equation 7—and confirms Equation 7 against the realized shifts of Appendix C at every event time in the window.

Data and code availability

This paper uses no data of any kind that are not reproduced in it. Every number it reports is produced either by a simulation that generates its own data from a stated seed or by

a deterministic algebraic computation on a stated design; the illustration of Section 8.1 is computed from month-in-sample indices published in Bailer (1975, Tab. 1), Solon (1986, Tab. 1) and McIllece (2022, Tab. 2), which are reproduced inside the code. No restricted or licensed microdata are used, and none are needed to reproduce any table or appendix of this paper.

The complete archive—the ten scripts, all of their outputs, the manuscript source and its build script—is public at <https://github.com/sokubo/paper-panel-conditioning-replication>. The fixed version corresponding to this manuscript is tag `paper-v0.7`, whose README gives the execution order, the environment, the runtimes, and the map from each table and appendix number to the script and output file that produce it. The commit of that tag checked against this manuscript is `a48bd80`; its `RELEASE_CHECK.md` records an anonymous download of the published snapshot, a comparison of its file list against the archive’s manifest, and a re-execution of the documented sequence in a clean copy with the regenerated outputs compared against the shipped ones. That record is the author’s own verification: no third party has re-run the archive.

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